

# ngspice-46 — Full Change Report

Every modification and its reason (original → current)

Ngspice + OpenVAF Enhancements project

July 2026

# Contents

|                                               |    |
|-----------------------------------------------|----|
| ngspice-46 — Full Change Report               | 3  |
| 1. The OSDI ABI, version 0.4 → 0.7            | 3  |
| 2. The OSDI runtime — src/osdi/               | 4  |
| 3. Analyses — src/spicelib/analysis/          | 5  |
| 4. Frontend — src/frontend/                   | 7  |
| 5. Parser and device registry — src/spicelib/ | 8  |
| 6. Maths — src/math/                          | 8  |
| 7. Build system                               | 10 |
| 8. Per-enhancement index                      | 10 |

## ngspice-46 — Full Change Report

Every modification applied to ngspice-46 in this project, with its reason. The baseline is the pristine ngspice-46 source tree (as vendored in the project's `original/` snapshot, which already contains OpenVAF-Reloaded's stock OSDI support); the current state is the tree committed in this repository under `ngspice-46/`. Each entry links the enhancement write-up in [enhancements\\_doc/](#) that carries the full engineering detail and the verifying example suite. The companion document [openvaf\\_changes\\_full-report.md](#) covers the compiler side.

Maintenance note: this report is updated whenever an enhancement touches ngspice sources. If you are reading it alongside a newer tree, the per-enhancement index at the end tells you the last change it covers.

Scope summary. One new source file and 41 modified ones carry all the functional changes (~2,000 substantive diff lines). A further ~100 `Makefile.in` files and `config.h.in` differ only because the auto-tools were regenerated with a current libtool ([E-77](#)); they contain no hand-written changes and are not listed individually. Everything below is behavior, interface, or diagnostics.

---

### 1. The OSDI ABI, version 0.4 → 0.7

`src/osdi/osdi.h` is the authoritative C contract between ngspice and the `.osdi` objects `openvaf-r` produces. The project extended it four times — twice additively (new exported symbols old loaders simply never look up), twice with genuine version bumps (layout changes):

| Change                                                                                                                       | Kind                                  | Enhancement          | Why                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>OSDI_ABSDELAY_COUNT</code><br>/<br><code>OSDI_ABSDELAY_INFO</code><br>exports                                          | additive<br>sym-<br>bols              | <a href="#">E-1</a>  | <code>absdelay()</code> needs simulator-side waveform history and synthetic delay-equation rows; the descriptors tell ngspice which nodes/offsets each delay slot uses                                                                                          |
| <code>OSDI_LAST_CROSSING_COUNT</code><br>/<br><code>OSDI_LAST_CROSSING_INFO</code><br>exports                                | additive<br>sym-<br>bols              | <a href="#">E-6</a>  | <code>last_crossing()</code> needs waveform history and crossing interpolation — same additive pattern as <code>absdelay</code>                                                                                                                                 |
| <code>OsdiNode.nodeset</code><br>field                                                                                       | ABI 0.5<br>(node<br>stride<br>change) | <a href="#">E-45</a> | Verilog-A net initializers ( <code>electrical a = 5.0;</code> ) become solver initial-guess nodesets; every node record carries the hint                                                                                                                        |
| <code>num_ac_stim_src</code><br>/<br><code>ac_stim_sources</code><br>/<br><code>load_ac_stim</code><br>descriptor tail       | ABI 0.6<br>(de-<br>scriptor<br>grows) | <a href="#">E-51</a> | full <code>ac_stim()</code> support: a Verilog-A module can <i>be</i> the AC stimulus, so the descriptor must enumerate its AC right-hand-side sources                                                                                                          |
| <code>load_noise</code><br>destination stride 1<br>→ 2 (signed<br>( <code>flat</code> , <code>react</code> )<br>power pairs) | ABI 0.7                               | <a href="#">E-54</a> | correct noise physics: operating-point-dependent factors and <code>ddt()</code> -shaped ( $j\omega$ ) noise need a complex amplitude per source, $re + j\omega \cdot im$ , so coherent same-named sources (LRM 4.6.4) sum with correct sign and frequency shape |

Two additive flag conventions ride on the existing `eval()` interface (not version bumps):

- `EVAL_FLAG_IS_FINAL_STEP` (bit 1 ≪ 21) in the `eval input` flags — `@(final_step)` firing ([E-53](#));
- `eval return` flags for `$finish/$stop` requests and `$discontinuity`-driven step rejection ([E-55](#)).

Pairing consequence: this ngspice requires  $\text{OSDI} \geq 0.7$  and rejects older objects with a clear version message; `.osdi` files from older compilers must be recompiled (E-54).

---

## 2. The OSDI runtime — `src/osdi/`

**osdiaccept.c** — new file (E-55, E-1, E-6)

The accepted-timepoint hook (`DEVaccept`) OSDI previously lacked. It advances the `absdelay/last_crossing` waveform histories only at *accepted* points (so rejected timesteps don't corrupt the delay lines) and reads the per-attempt latched `$finish/$stop` request flags at the one boundary where honoring them is safe. Why: acting on `$stop` mid-Newton-iteration was treated as a step failure and ground the timestep into a rejection loop; `$finish` was ignored outright.

**osdidefs.h** (104 diff lines)

- `OsdiExtraInstData` grows the per-instance fields behind every runtime feature: `dt/temp` offsets with given-flags (instance temperature), `eval_flags`, `absdelay` waveform-history rows and pre-allocated KLU/SPARSE matrix-pointer sets for the delay-equation rows (re-pointed on every DC↔AC transition, mirroring the regular Jacobian handling), and the `last_crossing` history (E-1, E-6, E-55).
- `ALIGN` macro renamed `OSDI_ALIGN`: the macOS SDK's `<sys/param.h>` defines an unrelated `ALIGN`, producing a redefinition warning in every OSDI translation unit (E-77).

**osdiitf.h / osdiext.h**

The registry-entry structure gains the `absdelay/last_crossing` descriptor pointers, and `OSDIpendingRequests()` is exported so the analyses can ask "did any device request `$finish/$stop` at this accepted point?" (E-1, E-6, E-55).

**osdiregistry.c** (68 diff lines)

- Loads the `absdelay/last_crossing` descriptor arrays from each `.osdi` and threads them into the registry entries (E-1, E-6).
- Requires  $\text{OSDI} \geq 0.7$  (E-54).

**osdiinit.c** (8 diff lines)

- Registers the `DEVaccept` hook (E-55).
- Publishes the synthetic instance-temperature parameter under both spellings, `dt` and the conventional `dtemp` every built-in device uses; `n1 a b mod dtemp=10` used to fail with "*unknown parameter*" while the underlying plumbing was exact (E-80).

**osdiload.c** (441 diff lines — the largest OSDI change)

- Stamps the `absdelay` synthetic rows (`y_synth`, `z`) for DC, AC and transient, driving the delay from the recorded history (E-1).
- Evaluates `last_crossing` from the recorded waveform with linear interpolation between the bracketing timepoints (E-6).
- Passes the final-step flag through to `eval()` (E-53).
- Latches `eval`-return flags per timepoint attempt (`point_eval_flags`, OR-ed across Newton iterations, reset per attempt) so `$finish/$stop` under solution-dependent conditions survive to the accepted-point boundary (E-55).

- Folds the stride-2 signed noise-power pairs ( $\text{fac} \cdot |\text{fac}|$  semantics) when transferring `load_noise` results (E-54, E-42).

#### **osdisetup.c** (178 diff lines)

- Creates the synthetic delay-equation unknowns and matrix entries for `absdelay` (E-1).
- Applies the OSDI nodesets (`OsdNode.nodeset`) as solver initial guesses (E-45).
- A `$fatal/$finish` raised during setup (models rejecting their configuration by design — HiSIM's `$port_connected` guards) used to surface as ngspice's baffling *"impossible error — can't occur"*; it now reads *"a Verilog-A device rejected its configuration during setup"* (E-56).
- An internal OSDI node that appears in no Jacobian entry — a net structurally decoupled from the matrix, e.g. an explicit ground `gnd` reference whose branch contribution `V(p,gnd) <+ ...` drops the `gnd` column — used to be allocated its own solver row. That all-zero row/column is benign under Sparse (it decouples to  $V=0$ ) but makes the KLU matrix structurally singular, so KLU returned a wrong DC answer (groundcontrib:  $v(p)=0$  not 1.5; hierbranch: branch currents 0). Such nodes are now tied to ground (node 0) at setup, matching how OpenVAF already treats them and fixing both solvers (E-116).

#### **osdiacld.c** (89 diff lines)

- AC loading gains the `ac_stim` right-hand-side injection: the partitioned AC-stimulus source array is loaded through `load_ac_stim` and added to the AC RHS with exact magnitude and phase, `$mfactor` applied linearly (deterministic signals scale with multiplicity) (E-51, E-26).
- Complex ( $j\omega$ -shaped) noise coupling entries participate in the AC matrix (E-54).

#### **osdinoise.c** (93 diff lines)

- Per-instance grouping of same-named noise sources: coherent summation of complex amplitudes  $(a + j\omega \cdot b) \cdot T$  before squaring, per LRM 4.6.4 — same-named sources correlate (including exact anti-phase cancellation), differently-named ones stay independent (E-42).
- Reads the stride-2 signed pairs of ABI 0.7 (E-54).

#### **osditrunc.c** (34 diff lines)

- Honors `$discontinuity(n)`'s next-step clamp: a negative `bound_step` sentinel from the model limits the step after the event (E-24).
- Honors the step-rejection return flag: when an event fired inside the step, `OSDitrunc` requests `delta/8` (with a  $20 \cdot \text{CKTdelmin}$  termination floor) so the integrator bisects onto the discontinuity instead of extrapolating across it (E-55).

#### **osdi/Makefile.am**

Adds `osdiaccept.c` to the build (E-55).

### 3. Analyses — **src/spicelib/analysis/**

#### **dctran.c** (46 diff lines)

- Advances OSDI delay/crossing histories via the accept path and skips history corruption on rejected steps (E-1, E-6).
- Issues the dedicated `@(final_step)` evaluation at the converged end of a successful transient (E-53).

- Honors accepted-point `$finish` (ends the analysis cleanly, firing `final_step` at the requesting point), `$stop` (pauses resumably), and aborts with a clear error on `$fatal`'s `E_PANIC` instead of retrying it as nonconvergence ([E-55](#)).

#### **dcop.c** (9) and **acan.c** (10)

- Fire `@(final_step)` at their successful end (an operating point fires both step events — a single point is first *and* last) ([E-53](#)).
- An AC job's operating-point phase carries the analysis name bit (only the name — adding the reactive `CALC_*` bits would wrongly enable integration at an op), so `analysis("ac")` holds through the whole AC run per LRM 4.6.1 and `@(initial_step("ac"))` fires in AC ([E-53](#)).

#### **dctrcurv.c** (150) + **include/ngspice/trcvdefs.h** (13)

- Generic `.dc @inst[param]` sweeps (a new sweep code): the DC sweep previously hardcoded `Vsource/Isource/Resistor/temperature`, so sweeping any other device parameter was a fatal error. The generic sweep resolves `@inst[param]` through the device's own parameter tables, refreshes per point via `DEVtemperature` (for OSDI exactly the `alter` path), nests with other sweep variables, and restores the original value afterwards ([E-62](#)).
- Fires `final_step` at the last sweep point; honors a `$finish` raised mid-sweep ([E-53](#), [E-55](#)).

#### **noisean.c** (37)

- A singular AC matrix during noise analysis crashed ngspice with a SIGABRT: the code ignored `NIacIter`'s return and the adjoint solve asserted on the unfactored matrix. It now aborts cleanly ("AC solution failed at ... Hz") ([E-56](#)).
- Honors deferred `$finish/$stop` raised at the noise operating point and fires `final_step` on success ([E-55](#), [E-53](#)).

#### **span.c** (31) + **cktspdum.c** (14)

- 1-port S-parameter analyses enabled: the "we need at least two ports" error was over-strict (a 1-port is a plain reflection measurement; the matrix machinery is N-general) — and it was hiding the `cadjoint` crash fixed in `dense.c` ([E-64](#)).
- `Rbase` published into the `sp` plot from port 1's `z0`, making the Touchstone writers work with no manual `let Rbase = 50` incantation ([E-64](#)).
- Stock NaN fixed in `donoise` noise-parameter extraction: the uncorrelated noise conductance `Gu` is analytically zero for fully-correlated single-source topologies, and floating-point rounding could land `sqrt(Ycor2 + Gu/Rn)`'s argument at  $-10^{-18} \rightarrow \text{NaN}$  for the noise figure. Clamped to the physical range  $\geq 0$  — found by parity testing against an OSDI resistor that produced the exact textbook NF where the built-in returned NaN ([E-63](#)).

#### **cktdisto.c** (16)

- `.disto` silently reported zero distortion for OSDI devices — the distortion kernel needs higher-order Taylor coefficients the OSDI ABI cannot provide (first derivatives only), and such devices were skipped without a word. A prominent warning now names each affected OSDI device type ([E-62](#)).

#### 4. Frontend — `src/frontend/`

**plotting/pyplot.c** (new) + **com\_pyplot.c** (new) + **plotit.c, commands.c** (E-94)

A new interactive command, **pyplot**, plots simulated vectors with matplotlib — a Python counterpart to gnuplot. `ft_pyplot()` (`plotting/pyplot.c`) mirrors `ft_gnuplot()`: it writes a `<file>.data` table and a `<file>.py` matplotlib script and `system()`s `python3` (synchronously for a PNG, backgrounded for an interactive window). A new "pyplot" device arm in `plotit()` routes to it, so it accepts the same plot expressions as `plot/gnuplot`; the `com_pyplot()` wrapper mirrors `com_gnuplot()`, and `pyplot` is registered in both command tables. Two set variables tune it: `pyplot_terminal=png` renders headless (Agg) to a PNG, and `pyplot_python` picks the interpreter (default `python3`). Purely additive (E-94). The output file base name is optional (E-95): `com_pyplot()` treats the first word as a file name only when it is not a plot expression (no `(`, and not a vector by `vec_get()`), otherwise it defaults to `pyplot` — so `pyplot v(out)` plots directly (the command's minimum argument count was lowered from 2 to 1). `ft_pyplot()` also grew stacked subplots (set `pyplot_subplots=N` → `plt.subplots(nrows, 1, sharex=True)`, `N` traces per panel; `vs` stays the x-axis, so it is not a panel separator) and matplotlib style sheets (set `pyplot_style=<name>`, `dark` → `dark_background`, guarded) (E-98). The headless `pyplot_terminal` was extended from PNG-only to the vector export formats `svg` and `pdf` (set `pyplot_terminal=svg/pdf` → `fig.savefig(<file>.<fmt>)`, matplotlib picking the writer from the extension), and a figure size control was added (set `pyplot_figsize="W,H"` → `plt.subplots(..., figsize=(W, H))`; quote the value so ngspice keeps the comma) (E-99).

**postcoms.c** (414 diff lines) + **postcoms.h** + **commands.c** (16)

The Touchstone I/O subsystem (E-64, E-72):

- **wrsnp** (with `wrs2p` dispatching to the same handler): Touchstone v1 for any port count — the classic 2-port `S11 S21 S12 S22` column order preserved byte-identically,  $N \geq 3$  in the spec's row-major layout with at most four complex pairs per line;
- writer options `wrsnp <file> [ri|ma|db] [s|y|z] [hz|khz|mhz|ghz]`, any combination in any order: MA/DB formats, Y/Z parameters (normalized to `Rbase`,  $Y \cdot R / Z / R$ , per the v1 spec), frequency units — the option line reflects every choice;
- **rdnsnp**: reads any Touchstone v1 file into a new plot with a Hz frequency scale and complex vectors matching the `.sp` plot's conventions (MA/DB converted back, Y/Z de-normalized, the 2-port column order handled, `Rbase` published so imports round-trip) — measured data diffs against simulation in one `let` expression.

Why: E-63 showed the S-parameter *analysis* was exact but its results could not leave ngspice in the industry-standard format (`wrs2p` demanded a never-created `Rbase` vector), and nothing could bring VNA data in.

**outitf.c** (16)

- Integer operating-point variables recorded per point: `getSpecial` masked with `IF_REAL` only, so an integer opvar saved with `.save @n1[flag]` produced garbage in transient vectors (E-32).
- The plot-memory guard's message printed `%Id` — a Windows-only printf length modifier that is undefined behavior elsewhere and rendered the message as the numberless "memory required (Id Bytes)". Now `%zu`: real byte counts on every platform (E-77).

**resource.c** (27)

`ft_ckptspace()` — the "approaching max data size" warning (`RSS > 95%` of `RSS` + free RAM) — now honors `set no_mem_check` (the same opt-out the fatal output-size estimate in `outitf.c` respects; one variable governs both memory guards) and warns once per excursion above the threshold, re-arming below 90%, instead of repeating on every check through a long analysis (E-81).

## **display.c** (1)

#undef NODEV before the local macro definition — the macOS SDK's <sys/param.h> defines an unrelated NODEV ([E-77](#)).

---

## 5. Parser and device registry — **src/spicelib/**

### **parser/ingtok.c** (10)

A **.model** card override silently dropped its first parameter when the type name and the opening parenthesis had no space between them (`modname devtype(p1=v1 ...)`): INPgetNetTok's scan did not treat ( as a token boundary, so the type token swallowed the paren and the first parameter name. Found while verifying event-function counters whose model-card overrides mysteriously kept defaults ([E-8](#)).

### **parser/ingmod.c** (5)

`find_model_parameter()` dereferenced `*(device->numModelParms)`, which is NULL for built-ins that take no model cards (VCVS, CCCS, ...) — any *referenced* `.model m vcvs()` card segfaulted, with no OSDI involved at all (an ordinary MOS instance sufficed). This was the true root of the long-documented "module named like a built-in crashes" gotcha. One NULL guard; both shapes now produce clean, located errors ([E-76](#)).

### **devices/dev.c** (51)

- Duplicate device-type registration warned and skipped: `osdi_add_device()` appended every descriptor unconditionally, and the model-card lookup scans front-to-back — so a module name duplicated across two loaded libraries silently resolved to the *first* registration (loading an updated model library gave the stale device with no hint), and a module named like a built-in corrupted model creation. Registration now checks the table case-insensitively and skips duplicates loudly ([E-76](#)).
- **pre\_osdi** path dedup: loading an already-loaded file notes it and skips — with the remedy stated: *"restart ngspice to load a recompiled file"* ([E-76](#), [E-81](#)).

### **osdi/osdiparam.c** (E-93)

Setting a fixed (**localparam**) OSDI parameter from the netlist used to be swallowed silently: `openvaf` exports every `localparam` as a parameter, its "given" flag is forced false, and the written value is overwritten by the parameter-init default — so `.model ... N=8` on a frozen structural width parameter changed nothing, with no feedback. `OSDIparam/OSDIiParam` now test the new `PARAM_FLAG_FIXED` descriptor flag (set by `openvaf`, a free bit of the parameter `flags` field) and emit *"parameter 'N' is a fixed (localparam) value and cannot be set from the netlist; ignored"* instead of silently dropping it ([E-93](#)).

---

## 6. Maths — **src/maths/**

**dense/dense.c** (11 substantive lines; the file also carries a whitespace/line-ending normalization from editing)

**cadjoint()** had no  $1 \times 1$  base case: its cofactor loop allocated  $0 \times 0$  and then negative-sized minors, killing ngspice with `malloc: can't allocate -8 bytes` — the crash that had been hiding behind the "at least two ports" guard in `span.c`. The adjugate of `[a]` is `[1]` (the empty minor's determinant is 1 by convention), making `cinverse` of a  $1 \times 1$  equal  $1/a$ ; a  $100\ \Omega$  load on a  $50\ \Omega$  port now writes a proper `.s1p` with  $S_{11} = 1/3$  exactly ([E-64](#)).

### **sparse/spfactor.c** (6)

The Markowitz-product overflow guards compare a double against `LARGEST_LONG_INTEGER` (= `LONG_MAX`), which is not exactly representable as a double; the conversion is now explicit — identical semantics, intentional and warning-free ([E-77](#)).

### **KLU/klu\_multiply.c** (16)

`SMPmultiply`'s KLU path passed `NULL` internal↔external ordering maps to `KLU_matrix_vector_multiply`, which dereferenced them unconditionally (`&NULL[n] = 0x14` for `n = 5`) and segfaulted — so `.option linesearch` crashed under `.option klu`. The line search is the only caller of `SMPmultiply` under KLU, so this path had never been exercised. A `NULL` map now means the identity ordering (`ext = i + 1`, since the KLU CSR is 0-based and the RHS/Solution vectors are 1-based), making the KLU matrix-vector product — and hence the [E-111](#) line-search residual merit — correct under KLU, with a merit sequence numerically identical to Sparse 1.3 ([E-112](#)).

### **KLU/klusmp.c** (SMPcaSolve)

`SMPcaSolve` is the complex adjoint (transposed) solve used by noise and S-parameters. Its Sparse branch calls `spSolveTransposed`, but the KLU branch called the *non-transposed* `klu_z_solve` — silently wrong for any asymmetric matrix (every circuit with a transistor or controlled source), which is why `.noise` was disabled under KLU. It now calls `klu_z_tsolve` (transposed), so KLU noise matches Sparse exactly (including OSDI device models). This is the core of [E-113](#), which also removes the KLU refusal guards in `noisean.c` / `pzan.c` (single-ended pole-zero runs under KLU; balanced-output pole-zero keeps a targeted guard).

### **spicelib/analysis/cktsens.c**

Sensitivity builds an auxiliary perturbation matrix `delta_Y` (holding  $\partial Y / \partial p$ ) via `SMPnewMatrix`, which allocates a plain Sparse 1.3 matrix (`delta_Y->CKTkluMODE = 0`, no `SMPkluMatrix`). It is only *multiplied* against the solution (`SMPmultiply` → `spMultiply`), never factored, and `DEVbindCSC` always binds into `ckt->CKTmatrix`, not `delta_Y` — so it is correctly Sparse in every case. But two KLU-only setup blocks were gated on the main matrix's `CKTkluMODE`, so under `.option klu` they ran and dereferenced the `NULL` `delta_Y->SMPkluMatrix` — a segfault on every DC/AC `.sens` deck. The blocks now gate on `delta_Y->CKTkluMODE` (its own flag, `0`), keeping `delta_Y` Sparse under KLU exactly as it already is under the default solver, while the main `Y` matrix stays KLU. DC and AC sensitivity now match Sparse exactly ([E-114](#)).

### **spicelib/analysis/distoan.c**

Distortion is a complex analysis (Volterra series solved at the harmonic / intermodulation frequencies via `NIIdIter` → `SMPcSolve`), but `distoan.c` had no KLU code at all: under `.option klu` the KLU matrix stayed in *real* mode, the complex solves ran against an unconverted matrix, and every harmonic came back zero — a silent wrong answer. The fix mirrors `acan.c`: before the solve loop it converts the matrix to complex (`DEVbindCSCComplex` per device + `KLUmatrixIsComplex`), and on exit converts back to real (`DEVbindCSCComplexToReal`) so a later real analysis in the same session is unaffected. KLU distortion now matches Sparse bit-for-bit (single-tone, two-tone intermodulation, and OSDI models) ([E-115](#)).

## 7. Build system

### xspice/icm/makedefs.in (4)

The XSPICE codemodel (.cm) link rule hardcoded the pre-macOS-10.3 idiom `-bundle -flat_namespace -undefined suppress`, deprecated loudly by the modern linker on all 14 codemodel links (CI macOS builds included). Now `-bundle -undefined dynamic_lookup` — the modern spelling for plugins whose host symbols resolve at load time (E-77).

### Autotools regeneration (the ~100 **Makefile.in** files + **config.h.in**)

Regenerated by `./autogen.sh` with libtool 2.5.4 during the zero-warning work — a build tree whose configure predates libtool 2.4.7 selects the deprecated darwin `-undefined suppress` flag whenever `MACOSX_DEPLOYMENT_TARGET` is unset. No hand-written content (E-77).

## 8. Per-enhancement index

Every enhancement that touched ngspice, oldest first:

| Enhancement | ngspice files                                                                                         | One line                                                                                  |
|-------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| E-1         | osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c, osdisetup.c, dctran.c (+accept path)       | <code>absdelay()</code> : synthetic delay rows + waveform history                         |
| E-6         | osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c                                             | <code>last_crossing()</code> : history + interpolated crossings                           |
| E-8         | parser/ingptok.c                                                                                      | .model card first-parameter drop fix                                                      |
| E-24        | osditrunc.c                                                                                           | <code>\$discontinuity</code> next-step clamp                                              |
| E-25        | osdi callbacks (get_simparams)                                                                        | expose <code>analysis_name</code> + simulator <code>simparams</code>                      |
| E-32        | outitf.c                                                                                              | integer opvars recorded per point                                                         |
| E-42        | osdinoise.c                                                                                           | coherent same-named noise grouping (LRM 4.6.4)                                            |
| E-45        | osdi.h (ABI 0.5), osdisetup.c                                                                         | Verilog-A nodesets applied to the solver                                                  |
| E-51        | osdi.h (ABI 0.6), osdiacld.c                                                                          | full <code>ac_stim</code> AC-RHS injection                                                |
| E-53        | dctran.c, dcop.c, dctrcurv.c, acan.c, noisean.c, osdiload.c                                           | <code>@(final_step)</code> + analysis-name phase bits                                     |
| E-54        | osdi.h (ABI 0.7), osdinoise.c, osdiload.c, osdiacld.c, osdiregistry.c                                 | node-free complex noise factors, stride-2 pairs                                           |
| E-55        | osdiaccept.c (new), osdiload.c, osditrunc.c, osdiinit.c, dctran.c, dctrcurv.c, osdiitf.h, Makefile.am | <code>\$finish/\$stop/\$fatal</code> honored; <code>\$discontinuity</code> step rejection |
| E-56        | noisean.c, osdisetup.c                                                                                | noise SIGABRT fix; setup-rejection diagnostics                                            |
| E-62        | dctrcurv.c, trcvdefs.h, cktdisto.c                                                                    | generic .dc @inst[param] sweeps; .disto warning                                           |
| E-63        | span.c                                                                                                | donoise NaN clamp (stock defect, found by parity)                                         |
| E-64        | span.c, cktspdum.c, dense.c, postcoms.c/.h, commands.c                                                | Touchstone export, auto-Rbase, 1-port .sp, cadjoint 1x1                                   |
| E-72        | postcoms.c/.h, commands.c                                                                             | Touchstone round 2: MA/DB/Y/Z/units + rdsnp reader                                        |

| Enhancement | messspice files                                                                        | One line                                                                                                                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-76        | dev.c, inpgmod.c                                                                       | duplicate-registration warning, load dedup, stock .model segfault fix                                                                                                                                                                     |
| E-77        | osdidefs.h, display.c, outitf.c, spfactor.c, makedefs.in (+autotools regen)            | zero-warning build, 33 → 0                                                                                                                                                                                                                |
| E-80        | osdiinit.c                                                                             | dtemp instance-parameter alias                                                                                                                                                                                                            |
| E-81        | resource.c, dev.c                                                                      | memory-warning opt-out + once-per-excursion; reload-note hint                                                                                                                                                                             |
| E-93        | osdi.h, osdiparam.c                                                                    | warn (not silently ignore) when a netlist sets a PARA_FLAG_FIXED (localparam) OSDI parameter                                                                                                                                              |
| E-94        | plotting/pyplot.c (new), com_pyplot.c (new), plotit.c, commands.c                      | new pyplot command — plot vectors with matplotlib (a Python gnuplot)                                                                                                                                                                      |
| E-95        | com_pyplot.c, commands.c                                                               | make the pyplot output file name optional (defaults to pyplot)                                                                                                                                                                            |
| E-98        | plotting/pyplot.c                                                                      | pyplot stacked subplots (pyplot_subplots=N) + matplotlib style sheets (pyplot_style)                                                                                                                                                      |
| E-99        | plotting/pyplot.c                                                                      | pyplot vector export formats (pyplot_terminal=svg/pdf) + figure size (pyplot_figsize)                                                                                                                                                     |
| E-110       | optdefs.h, tsdefs.h, cktsopt.c, cktntask.c                                             | .option errpre-set=conservative moderate liberal — one knob for a coordinated tolerance/robustness set; explicit options override regardless of order (moderate = historical defaults)                                                    |
| E-111       | optdefs.h, tsdefs.h, cktdefs.h, cktsopt.c, cktdjob.c, cktntask.c, cktdtest.c, niiter.c | .option linesearch — globalized (damped) Newton via Armijo backtracking on a new KCL-residual merit $F = G \cdot x - b$ ; result-neutral, off by default                                                                                  |
| E-112       | maths/KLU/klu_multiply.c                                                               | KLU support for .option linesearch — SMPmultiply's KLU path passed NULL ordering maps that klu_matrix_vector_multiply dereferenced (SIGSEGV); NULL now means identity ordering. Line search verified merit-identical KLU vs Sparse        |
| E-113       | maths/KLU/klusmp.c, spicelib/analysis/noisean.c, pzan.c                                | KLU support for noise + single-ended pole-zero — SMPcaSolve's adjoint KLU branch used the non-transposed solve (silently wrong noise on asymmetric matrices); now klu_z_tsolve. Guards removed; balanced-output pz keeps a targeted guard |

| Enhancement | spice files                                             | One line                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-114       | spicelib/analysis/cktsens.c                             | KLU support for sensitivity — the auxiliary perturbation matrix <code>delta_Y</code> is Sparse, but two KLU setup blocks gated on the <i>main</i> matrix's flag dereferenced its NULL <code>SMPkluMatrix</code> (segfault on every DC/AC <code>.sens</code> ); now gated on <code>delta_Y</code> 's own flag. DC/AC <code>.sens</code> match Sparse exactly                                                               |
| E-115       | spicelib/analysis/distoan.c                             | KLU support for distortion — <code>.disto</code> is a complex analysis but <code>distoan.c</code> had no KLU code, so under KLU the matrix stayed real and every harmonic came back zero (silent wrong answer); now converts real↔complex around the solve loop like <code>acan.c</code> . Matches Sparse bit-for-bit                                                                                                     |
| E-116       | osdi/osdisetup.c                                        | KLU wrong-DC fix — an OSDI internal node with no Jacobian entry (a decoupled ground reference) was given an all-zero solver row that made the KLU matrix structurally singular; now tied to ground. Fixes the <code>groundcontrib</code> and <code>hierbranch</code> KLU_XFAILs                                                                                                                                           |
| E-117       | configure.ac (+configure),<br>spicelib/analysis/dcpss.c | Periodic steady state (PSS) productionized — built by default (was experimental <code>--enable-pss</code> , so <code>.pss</code> was unimplemented in shipped builds); ~230 lines of shooting-loop trace routed through <code>setngdebug</code> (232 → 31 lines by default); fail-fast KLU guard (PSS hangs under KLU) directing <code>.pss</code> to <code>.option sparse</code>                                         |
| E-118       | spicelib/analysis/dcpss.c                               | PSS runs under KLU — the shooting loop hung under KLU ( <code>klu_refactor</code> 's reused pivots inflated the truncation error into a ~20M-step timestep explosion); forcing a full re-factor (NISHOULDREORDER) each PSS step makes KLU converge to the same result as Sparse. E-117's Sparse-only guard removed                                                                                                        |
| E-119       | pssdefs.h, spicelib/analysis/dcpss.c                    | Retain the PSS periodic operating point — PSS sampled the node voltages per period for its DFT then freed them, and never captured device states; now the voltages and <code>CKTstate0</code> states are captured per sample and retained on the PSSan job (with frequency + dims) as the substrate for the periodic small-signal suite (PAC/pnoise/PXF). Self-checks that the retained samples reproduce the fundamental |

| Enhancement | ngspice files                                                                                                         | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-120       | spicelib/analysis/dcpss.c                                                                                             | Periodic small-signal Jacobian harmonics — walk the retained operating point, re-linearize each sample (CKTload MODEINITSMSIG) and stamp $G+jC$ (CKTload $\omega=1$ ), read the osc-node diagonal $\rightarrow g(t), c(t)$ , DFT to harmonics. The blocks the PAC conversion matrix is built from. (Fixed a complex-mode-not-set bug where $C(t)$ accumulated across samples.) Verified: RC gives $G=1/R1$ , $C=C1$ , no harmonics                                                                                                                                                                              |
| E-121       | spicelib/analysis/dcpss.c                                                                                             | PAC conversion-matrix engine — extend E-120 from the osc diagonal to every matrix nonzero, complex-DFT to harmonics $G_k$ , $C_k$ , assemble the $(2M+1)N$ harmonic conversion matrix $H_{\{nm\}}=G_{\{n-m\}}+j\omega_m \cdot C_{\{n-m\}}$ , inject a unit current at the osc node in sideband 0 and solve by dense complex LU (pss_solve), report the sideband conversion gains. The numerical heart of PAC/pnoise/PXF. Verified: linear RC gives sideband-0 = AC driving-point ‘                                                                                                                              |
| E-122       | include/ngspice/pssdefs.h,<br>spicelib/analysis/psssetp.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/dcpss.c | .pac command (periodic AC) — the user-facing analysis on the E-121 engine.<br>.pac Fguess StabTime OscNode Points Harmonics SC_iter Steady_coeff<br><dec oct lin> Npts Fstart Fstop runs PSS then sweeps the input frequency, solving the conversion matrix at each point and emitting the 0-th-sideband node responses as a complex PAC Analysis plot (print/plot/wrdata). Reuses the PSS analysis via a PSSdoPAC flag; engine factored into pac_extract_harmonics (once) + pac_solve_at (per freq). Verified: swept sideband-0 $ b(f)  ==$ analytic AC driving-point $ Z(f) $ to $1.6e-7$ across 10 kHz–1 MHz |

| Enhancement | ngspice files                                                                                                         | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-123       | include/ngspice/pssdefs.h,<br>spicelib/analysis/psssetp.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/dcpss.c | finish .pac — (1) source-referenced stimulus: capture the netlist AC-source RHS from CKTAcLoad as the sideband-0 stimulus B_0 (a periodic-AC transfer/conversion gain), unit-current-at-osc-node fallback; (2) multi-sideband output: optional trailing maxsideband Ksb emits every conversion sideband $f_{in+k} \cdot f_0$ ( $k=-Ksb..Ksb$ ) as its own vector — sideband 0 keeps the node name, others <code>&lt;node&gt;_usb&lt;k&gt;/&lt;node&gt;_lsb&lt;k&gt;</code> via <code>IFnewUId</code> . Verified: RC with V1 AC 1 gives sideband-0 = low-pass transfer $1/\sqrt{1+(2\pi fRC)^2}$ (0.998→0.157) with <code>b_usb1/b_lsb1 ~0</code> (no conversion)                                                                                                                      |
| E-124       | include/ngspice/pssdefs.h,<br>spicelib/analysis/psssetp.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/dcpss.c | .pnoise command (periodic noise) — fold each device's noise through the conversion matrix. <code>pac_solve_adjoint</code> solves $H^T \Psi = e_{\{out,0\}}$ ; <code>pnoise_sweep</code> loads the sideband-k adjoint into <code>CKTrhs/CKTirhs</code> and calls the existing device noise routines ( <code>NevalSrc</code> , <code>OSDI load_noise</code> ) once per sideband, accumulating $\Sigma_k S \cdot  \Delta\Psi_k ^2$ — reuses every device noise model via a minimal local <code>NOISEAN</code> context. .pnoise <code>&lt;pss&gt; OutNode InSrc &lt;dec oct lin&gt; Np Fstart Fstop</code> ; emits <code>onoise_spectrum/inoise_spectrum</code> . Verified: linear RC reduces exactly to .noise ( $4kTR/(1+(2\pi fRC)^2)$ ), matches the .noise reference to every digit) |
| E-125       | include/ngspice/pssdefs.h,<br>spicelib/analysis/psssetp.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/dcpss.c | .pxf command (periodic transfer function) — the adjoint of PAC, completing the <code>PSS→PAC→Pnoise→PXF</code> suite. <code>pxf_sweep</code> solves $H^T \Psi = e_{\{out,0\}}$ per frequency and dots each sideband block with the AC-source pattern $B_0 \rightarrow xf_k = \Sigma_j \Psi_k(j) \cdot B_0(j)$ , the input→output transfer at each sideband. .pxf <code>&lt;pss&gt; OutNode &lt;dec oct lin&gt; Np Fstart Fstop [maxsb]</code> ; emits <code>xf + xf_usb&lt;k&gt;/xf_lsb&lt;k&gt;</code> . Verified: by the identity $(H^{-1}B)_{out} = (H^{-T}e_{out})^T B$ the sideband-0 transfer is bit-identical to the PAC response (0.998→0.157 low-pass), conversion sidebands $\sim 2e-16$                                                                                    |

| Enhancement | ngspice files                                                                                                                     | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-126       | include/ngspice/pssdefs.h,<br>spicelib/analysis/psssetp.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/dcpss.c             | cyclostationary noise — <code>.pnoise ... cyclo</code><br>evaluates each device's noise PSD $S(t)$ at every PSS sample's bias (CKTload per sample) and folds it through the time-domain adjoint transfer $A_s(j) = \sum_k \Psi_k(j) e^{j2\pi k s/P}$ , averaging: $\text{onoise}(f) = (1/P) \sum_s S(t_s) \cdot  \Delta A_s ^2$ . Reuses the device noise routines. Verified: (1) reduces exactly to <code>.noise</code> on the linear RC (bias-independent thermal $\rightarrow$ Parseval); (2) a flicker resistor carrying the RC current gives $\text{onoise} \cdot f = R1^2 \cdot KF \cdot \langle I^2 \rangle = 4.88e-10$ (uses the period-average $\langle I^2 \rangle$ , matched to 5 digits)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| E-127       | include/ngspice/{optdefs,tskdefs,cktdefs}.h,<br>spicelib/analysis/{cktsopt,cktnntask,cktdojob,cktdontest}.c,<br>maths/ni/niiter.c | pseudo-transient continuation — <code>.option pntc</code><br>continues $f(x)=0$ in a backward-Euler pseudo-transient $f(x) + G_{ps} \cdot (x - x_{\text{prev}}) = 0$ and marches $d\tau$ small $\rightarrow$ large ( $G_{ps} \rightarrow 0$ ). $G_{ps}$ diagonal via the <code>gmin</code> path; the $G_{ps} \cdot x_{\text{prev}}$ RHS coupling (added in <code>NIiter</code> ) makes each step a stable-trajectory move, not a static <code>gmin</code> step. New <code>pseudo_transient</code> in the CKTop cascade, off by default. Verified under KLU+Sparse: result-neutral on normal circuits; on a stiff no-limiting exp, reaches the correct DC $0.837922$ V where plain Newton lands on a spurious $70.5$ V (21/21)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| E-128       | include/ngspice/{optdefs,tskdefs,cktdefs}.h,<br>spicelib/analysis/{cktsopt,cktnntask,cktdojob}.c,<br>spicelib/analysis/dctran.c   | LTE-based dynamic integration-order control — <code>.option dynorder</code> selects the Gear order per step from the local-truncation-error limit instead of the stock $1 \leftrightarrow 2$ toggle, so orders 3–6 (already coded in <code>NIcomCof</code> but never used) are actually exercised. The selector compares the raw LTE-limited step (uncapped CKTtrunc) at the current order and its $\pm 1$ neighbours and moves with hysteresis ( $1.2\times$ ), a settling hold after each change (let the BDF divided differences rebuild), and an order-dependent step-growth cap ( $2\times$ at order $\leq 3 \rightarrow 1.3\times$ at 6). Off by default; bounded by <code>maxord</code> ; inert at the default <code>maxord=2</code> . Stiff-transient guards (post-breakpoint order hold + leaky-bucket rejection-rate order drop) keep it from collapsing the step on a violent slew. Verified under KLU+Sparse: RC decay $3\text{--}5\times$ fewer steps at matched accuracy with monotone error; smooth RLC ringdown $8.9\times$ fewer steps AND more accurate (0.13 % vs 0.34 %); nonlinear rectifier matches stock to 5 sig figs; the transistor-level $\mu A741 \pm 5$ V slew completes under <code>dynorder</code> and matches fixed Gear-2 |

| Enhancement | ngspice files                                                                                                                | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-129       | frontend/outitf.c                                                                                                            | <p>sweep progress bar — the throttled Reference value status line (redrawn in place every 0.25 s during a sweep) now carries a live progress bar + percentage, e.g. Reference value : 5.91926e-04 [=====] 74%.<br/> outp_progress_frac() computes the 0-1 fraction per analysis: transient from (CKTtime-TSTART)/(TSTOP-TSTART), AC/noise from the frequency's linear/log position in the band, DC from the accepted-point count over the nested step-count product; analyses with no span (op, ...) keep the plain line. Fixed-width (no stale chars), stdout status-line only (never in the rawfile/wrdata), same !ft_norefprint &amp;&amp; !cp_background gating. Verified 22/22: printed % matches the analytic sweep fraction to 0.5 % across tran/AC/DC/noise; op shows no bar</p>                           |
| E-130       | frontend/com_optimize.{c,h}, commands.c, options.c, include/ngspice/ftext.h, spicelib/analysis/cktdojob.c, frontend/outitf.c | <p>built-in Nelder-Mead optimizer — optimize -param &lt;name&gt; &lt;init&gt; &lt;lo&gt; &lt;hi&gt; [-param ...] -analysis &lt;cmd&gt; -minimize &lt;expr&gt; [-maxiter N] [-tol T] [-verbose] varies device/alter parameters, re-runs an analysis, and minimizes a scalar objective expression via a derivative-free downhill simplex in normalized [0,1] space (scale-invariant across orders-of-magnitude params). Sub-commands dispatched synchronously through cp_coms (not the deferring re-entrant cp_ev loop); the hundreds of inner analyses are silenced via a new ft_optimizing flag gating the banner/row-count/E-129 progress bar at source (-verbose to show). Verified 9/9 against analytic optima: DC divider R1→2333.3 Ω, AC low-pass R1→2756.6 Ω, 2-D compound objective R1=3k/R2=2k exactly</p> |

| Enhancement | ngspice files                                                                                                                 | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-131       | frontend/com_checkpoint.{c,h}, commands.c, com_commands.h, Makefile.am, spicelib/analysis/dctran.c, include/ngspice/cktdefs.h | transient checkpoint / restart — new savestate <file> / loadstate <file> commands serialize the full transient integration state (solution vector CKTrhsOld, device state history CKTstates [], time/step/order/mode, pending breakpoints) to a binary file and resume it, including in a fresh process — so a long run survives a crash, splits across sessions, or moves between machines (stock ngspice could only resume in memory). Dctran() gains a CKTcheckpoint-gated branch that opens a fresh output plot (no live plot to 666-relink across a reload), initializes the XSPICE breakpoint markers, and fixes up the breakpoint list before jumping into the stepping loop; the rhs length is keyed off SMPmatSize+1 (not CKTmaxEqNum). A stored signature rejects a mismatched circuit; Sparse-solver only (KLU rejected with a clear message). Verified 19/19: resumed waveform bit-identical to an uninterrupted run for RC-step/pulse/diode built-ins and ~2e-7 for a compiled OSDI diode, across a separate process |

| Enhancement | ngspice files                                                                                                         | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-132       | spicelib/analysis/dcpss.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/psssetp.c,<br>include/ngspice/pssdefs.h | periodic S-parameters (.psp) — small-signal scattering parameters around a PSS operating point, including conversion between the input frequency and its sidebands $f_{in+k} \cdot f_0$ (mixers / switched circuits, where a static-DC .sp cannot see the conversion). Sits on the PSS→conversion-matrix suite (E-117–126): after PSS, psp_sweep excites each RF port (portnum/z0, the .sp framework) by driving its branch source (V=1, like .sp's VSRCspupdate) through the shared $(2M+1)N$ conversion matrix, reads per-sideband port waves in the same Kurosawa power-wave convention, and forms $S^{(k)} = B^{(k)} \cdot A^{-1}$ (dense-complex cinverse/cmultiply). pac_solve_at's matrix assembly factored into a reusable pac_build_matrix; PSSdoPSP flag + psp PSS param + dot_psp card. Because $S = B \cdot A^{-1}$ is excitation-basis-invariant, sideband 0 reduces exactly to .sp for a time-invariant network. Runs under both linear solvers (the conversion matrix is a standalone dense LU; PSS runs under both since E-118). Verified 8/8: sideband-0 matches .sp to $\sim 1e-16$ for 1/2/3-port resistive + reactive networks (magnitude and phase) incl. OSDI Verilog-A devices (G + reactive ddt stamps), conversion sidebands correctly $\sim 0$ |

| Enhancement | ngspice files                                                    | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-133       | frontend/com_qpss.{c,h}, commands.c, com_commands.h, Makefile.am | <p>quasi-periodic (two-tone) steady state (qpss)</p> <p>— qpss &lt;expr&gt; &lt;f1&gt; &lt;f2&gt; [periods] [maxorder] computes the two-tone steady-state spectrum: every mixing product <math>k_1 \cdot f_1 + k_2 \cdot f_2</math> including third-order intermodulation (IM3 at <math>2f_1 - f_2</math> / <math>2f_2 - f_1</math>) that single-tone AC/PSS cannot show. For commensurate tones (common beat <math>fb = \gcd(f_1, f_2)</math>) it runs an ordinary transient over a few beat periods to reach steady state, then evaluates the Fourier coefficient directly at each exact intermod frequency (a direct DFT, exact — no FFT-bin rounding) and labels it by the 2-D index <math>(k_1, k_2)</math>.</p> <p>Deliberately not a slow beat-frequency shooting PSS; a front-end command driving a transient, so solver-independent and works with built-in and OSDI devices. Verified 11/11: analytic fundamentals/IM3/3f of a cubic, no even-order products, the 3:1 IP3 slope law (fund <math>\times 2</math>, IM3 <math>\times 8</math> per <math>2\times</math> drive), beat-frequency derivation, and an OSDI cubic matching the built-in</p> |

| Enhancement | ngspice files                                                                                                                 | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-134       | spicelib/analysis/dcpss.c,<br>frontend/com_hb.{c,h}, commands.c,<br>com_commands.h, Makefile.am,<br>include/ngspice/cktdefs.h | <p>Harmonic Balance (hb &lt;f0&gt; &lt;K&gt;) — solves the periodic steady state in the frequency domain by Newton (each node a truncated Fourier series), instead of integrating in time; the real analysis behind ngspice's unimplemented WITH_HB stub. Residual <math>F_k = I_{R,k}(V) + [dq/dt]_k - Is_k = 0</math> with the E-121 <math>(2K+1)N</math> conversion matrix as the exact Jacobian. Per iteration: inverse-DFT <math>V \rightarrow v(t_s)</math>; drive DC+AC device loads at those voltages for the resistive current <math>I_R = G*v - b</math> (companion b saved before acLoad, so it's the ACTUAL current not the tangent <math>G*v</math>), and <math>G(t)/C(t)</math>; DFT; dense complex Newton (pss_csolve). Nonlinear reactive with NO charge extraction: <math>dq/dt = C(v)*v'</math>, so the reactive current is the conversion matrix's <math>jwC</math> term applied to <math>V</math>. Solver-independent (KLU + Sparse): the dense complex Newton is HB's own; the sparse solver only <i>reads</i> <math>G(t)/C(t)</math> off the device matrix, so hb_extract carries the same <code>#ifdef KLU</code> complex-CSC binding (<code>DEVbindCSCComplex</code>) as the PAC extraction, and com_hb copies the task's KLU mode before building so a bare hb honours <code>.option klu</code> — verified bit-identical under both. Built-in + OSDI. Verified 8/8 vs transient/fourier, quadratic convergence: nonlinear R (analytic 3rd harmonic), R+C, a real diode rectifier (junction limiting), OSDI varactor <math>Q(v)</math> 2nd harmonic, KLU==Sparse parity</p> |

| Enhancement | ngspice files                                                   | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-135       | spicelib/analysis/dcpss.c,<br>examples/hb_examples/verify_hb.py | HB source-stepping continuation — makes E-134 Harmonic Balance robust on strongly-driven circuits where a cold full-strength Newton diverges ( $ F  \rightarrow 1e69$ ). HBanalyze's Newton loop is wrapped in an adaptive homotopy: every independent source scaled by $\lambda: 0 \rightarrow 1$ , each level warm-started from the last converged V. The residual becomes $F = I_R + I_C - \lambda \cdot I_s$ ; on level success V is checkpointed and $d\lambda$ grows ( $\times 1.7$ ), on failure (Newton exhausted, residual non-finite, or singular Jacobian) V is restored and $d\lambda$ halves; $d\lambda < 1e-5 \rightarrow$ reported as no reachable steady state. First level is full strength ( $d\lambda=1$ ) so easy circuits converge at $\lambda=1$ immediately, bit-identical to the plain solve. Automatic (no new syntax); set <code>hb_verbose</code> shows the $\lambda$ ramp, summary reports iterations + continuation steps. Verified 9/9: a strongly-driven diode rectifier (5 V into 20 $\Omega$ , sharp $I_S=1e-14$ ) diverges cold but converges in 3 continuation steps, matching transient fourier to $<0.1\%$ for DC/ $f_0/2f_0$ ; the 8 existing HB checks stay bit-identical |

| Enhancement | ngspice files                                                                                                                                     | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-136       | spicelib/analysis/dcpss.c,<br>frontend/com_qpss.c, frontend/commands.c,<br>include/ngspice/cktdefs.h,<br>examples/qpss_examples/verify_qpss_hb.py | two-tone Harmonic Balance (qpss <expr><br><f1> <f2> hb [K1] [K2]) — the TRUE,<br>incommensurate-capable quasi-periodic<br>steady state, a frequency-domain HB engine<br>alongside the E-133 transient qpss (which is<br>unchanged, commensurate-only). Each node<br>is a 2-D Fourier series $v(t) = \sum \sum V_{k1,k2} e^{j(k1\omega1+k2\omega2)t}$ ; QPSShb samples<br>devices on a 2-D phase grid ( $\theta1, \theta2$ ) — time<br>never appears, so incommensurate tones (no<br>beat period) just work — and 2-D DFTs to the<br>conversion matrix<br>$H_{(n),(m)} = G_{n-m} + j\omega_m \cdot C_{n-m}$ (qp_build_matrix), Newton-solved by<br>pss_csolve with the E-135 source stepping.<br>Sources captured by an oversampled<br>least-squares APFT ( $\Gamma^H \Gamma$ ) $I_s = \Gamma^H b$ (a<br>square Vandermonde is catastrophically<br>ill-conditioned past a few harmonics).<br>Reactive needs NO charge extraction<br>( $dq/dt = C(v)v'$ ); the converged operating<br>point is retained (qpss_hb_saved) for qpac<br>(E-137). Solver-independent (KLU + Sparse)<br>— same <code>#ifdef KLU</code> complex-CSC binding<br>as PAC/HB. Built-in + OSDI. Verified 7/7:<br>analytic cubic $ IM3(2,-1) / 3rd(3,0) =3$ ,<br>even products ~0, 3:1 IP3 slope,<br>incommensurate 2 tones (E-133 cannot),<br>HB==transient (commensurate),<br>KLU==Sparse; E-133 verify_qpss.py stays<br>11/11 |

| Enhancement | ngspice files                                                                                                                                                             | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-137       | spicelib/analysis/dcpss.c,<br>frontend/com_qpac.{c,h}, commands.c,<br>com_commands.h, Makefile.am,<br>include/ngspice/cktdefs.h,<br>examples/qpss_examples/verify_qpac.py | two-tone small-signal QPAC (qpac <f_in>)<br>— completes the quasi-periodic gap: the<br>two-tone analogue of PAC. Run after qpss ...<br>hb, QPACanalyze injects a small signal at<br>f_in around the retained QPSS operating<br>point (qpss_hb_saved) and reports the<br>response at every sideband<br>f_in+k1·f1+k2·f2, mixing it through the<br>SAME 2-D conversion matrix<br>$H_{\{(n), (m)\}} = G_{\{n-m\}} + j\omega_m \cdot C_{\{n-m\}}$ (qp_build_matrix at f_in) that the QPSS<br>Newton used as its Jacobian. Stimulus placed<br>in the (0,0) sideband — a netlist AC-source<br>RHS B0 (captured bias-independent at the<br>op-point) or a unit-current fallback — one<br>dense pss_csolve. Adds no device<br>evaluation → solver-independent (KLU +<br>Sparse) by construction. Verified 7/7:<br>reduce-to-AC (pump→0) direct (0,0) =<br>plain .ac response = R, sidebands vanish),<br>v <sup>2</sup> -pump conversion ratio<br>$ (1,1) / (2,0) =2$ , equal-tone symmetry,<br>clean no-op-point error, KLU==Sparse;<br>E-136 (7/7) + E-133 (11/11) unaffected |

| Enhancement | ngspice files                                                                                                                                                                   | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-138       | spicelib/analysis/dcpss.c,<br>frontend/com_qpnoise.{c,h}, commands.c,<br>com_commands.h, Makefile.am,<br>include/ngspice/cktdefs.h,<br>examples/qpss_examples/verify_qpnoise.py | two-tone small-signal QPnoise (qpnoise<br><output_node> <f_in>) — quasi-periodic<br>noise, the two-tone analogue of pnoise<br>(E-124), on the retained qpss ... hb operating<br>point. Reports output + input-referred noise<br>density at f_in, folding every device's noise<br>over all sidebands $f_{in} + k1 \cdot f1 + k2 \cdot f2$<br>(mixer/PA noise conversion invisible to a<br>static .noise). QPnoiseAnalyze biases the<br>devices at the QPSS operating point, then<br>qp_solve_adjoint transposes the 2-D<br>conversion matrix (qp_build_matrix) and<br>solves $H^T \Psi = e_{\{out, (0,0)\}}$ — one<br>adjoint gives the transimpedance from every<br>(node, sideband) to the output; each device's<br>DEVnoise computes $S \cdot  \Psi ^2$ (reading the<br>transfer from CKTrhs/CKTirhs) and the sum<br>over all Nh harmonics is the folded onoise;<br>input-referred via the QPAC gain. Reads only<br>retained data → solver-independent (KLU +<br>Sparse) by construction. With no pump the<br>conversion matrix is block-diagonal so only<br>(0,0) survives [?] reduces to .noise. Verified<br>6/6: reduce-to-noise (pump→0 [?] onoise ==<br>plain .noise == 4kTR exactly), conversion<br>active under pump, inoise=onoise/gain <sup>2</sup> ,<br>clean no-op-point error, KLU==Sparse;<br>E-133 (11/11) + E-136 (7/7) + E-137 (7/7)<br>unaffected |

| Enhancement | ngspice files                                                                                                                                          | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-139       | spicelib/analysis/dcpss.c,<br>frontend/com_qpnoise.c,<br>frontend/commands.c,<br>include/ngspice/cktdefs.h,<br>examples/qps_examples/verify_qpnoise.py | cyclostationary QPnoise (qpnoise<br><output_node> <f_in> cyclo) — upgrades<br>E-138 to a time-varying device PSD. Under a<br>two-tone pump a device's bias swings over<br>the period (a diode's shot noise $2qI_D(t)$<br>spikes when it conducts), so<br>QPnoiseAnalyze gains a cyclo branch using<br>the identity $onoise = (1/P) \cdot \sum_s$<br>$S(t_s) \cdot  A_s ^2$ , where $A_s(j) =$<br>$\sum_{\{(k1,k2)\}} \Psi_{\{(k1,k2)\}}(j) \cdot e^{j2\pi(k1$<br>$s1/P1+k2 s2/P2)}$ is the inverse 2-D DFT of<br>the adjoint transfers (the time-domain<br>transimpedance at phase sample s). Each<br>sample re-biases the devices at the retained<br>op-point (qp_synth, P1/P2 now stored in<br>qp_harm) with per-sample junction settling<br>(the E-134 MODEINITFLOAT walk — else a<br>limited diode reports a stale bias and the<br>"cyclostationary" result collapses onto the<br>stationary one), evaluates $S(t_s)$ , folds<br>through $A_s$ , and averages. By Parseval<br>reduces to the stationary sum (and .noise)<br>when S is constant. Verified 10/10 (6 E-138 +<br>4 cyclo): cyclo reduce-to-noise, Parseval<br>(bias-independent thermal PSD [?]<br>cyclo==stationary even under pump), a<br>hard-pumped diode where cyclo differs from<br>stationary by $\sim 8\times$ (switching-mixer<br>cyclostationary enhancement, visible only<br>with the settling), cyclo KLU==Sparse;<br>E-133/136/137 unaffected |

| Enhancement | ngspice files                                                                                                                                                                            | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-140       | spicelib/analysis/dcpss.c,<br>frontend/com_hbosc.{c,h}, commands.c,<br>com_commands.h, Makefile.am,<br>include/ngspice/cktdefs.h, exam-<br>ples/phasenoise_examples/verify_phasenoise.py | oscillator phase noise (hbosc <oscnod><br><K> [fguess] [tstab] + phasenoise<br><fstart> <fstop> [pts]) — closes the<br>periodic/phase-noise gap with the<br>phase-noise piece. HBOSCanalyze is an<br>autonomous harmonic balance: an oscillator<br>has no source, so $F(V)=I_R+[dq/dt]=0$ is<br>solved for the harmonics AND the unknown<br>oscillation frequency $\omega_0$ . The conversion<br>matrix $dF/dV=H$ is singular (right null space<br>= phase mode $u_k=j_k V_k$ , since the<br>oscillator's phase is free), so Newton runs on<br>the bordered system $[H, \partial F/\partial \omega_0; u^*, 0]$<br>(nonsingular by bordering, $\partial F/\partial \omega_0=I_C/\omega_0$ ,<br>gauge row $u^*$ ), transient-seeded (hbosc runs<br>a short transient from the deck's .ic, reads<br>amplitude + zero-crossing frequency),<br>converging quadratically (LC osc: 4 iters to<br>$ F =3e-12$ ; inductors fit the $G+j\omega C$ matrix via<br>branch-current MNA). PhaseNoiseAnalyze<br>builds the adjoint of $H$ at OFFSET df with the<br>unit at the carrier sideband ( $m=1$ ) and folds<br>device noise (DEVnoise); as $df \rightarrow 0$ the<br>limit-cycle matrix goes singular through the<br>phase mode so the transimpedance diverges<br>as $1/df \rightarrow$ folded noise $1/df^2$ , normalized to<br>carrier power $2 V_1 ^2 \rightarrow L(df)$ in dBc/Hz.<br>Verified 8/8 on an LC oscillator: autonomous<br>HB converges to $f_0$ +describing-function<br>amplitude, $L(df)$ has the $-20$ dB/dec skirt<br>near carrier flattening into the noise floor at a<br>physical level ( $\approx -147$ dBc/Hz @1kHz),<br>thermal scaling $L \propto T$ ( $2 \times T \rightarrow +3$ dB exactly,<br>pinning the absolute), clean no-op-point<br>error, KLU==Sparse; pnoise (9/9) + qpnoise<br>(10/10) unaffected |

| Enhancement | ngspice files                                                                                                                                                             | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-141       | spicelib/analysis/dcpss.c,<br>frontend/com_qpxf.{c,h}, commands.c,<br>com_commands.h, Makefile.am,<br>include/ngspice/cktdefs.h,<br>examples/qpss_examples/verify_qpxf.py | two-tone small-signal QPXF (qpxf<br><output_node> <f_in>) — the<br>quasi-periodic transfer function, the<br>ADJOINT of QPAC (E-137), completing the<br>two-tone small-signal suite<br>(QPSS→QPAC→QPnoise→QPXF ≡<br>PSS→PAC→pnoise→PXF). QPXFanalyze<br>runs one adjoint solve $H^T \Psi =$<br>$e_{\text{out}, (0,0)}$ (qp_solve_adjoint, reused<br>from QPnoise E-138) and dots each sideband<br>block of $\Psi$ with the netlist AC-source pattern<br>B0 (the QPAC stimulus, captured at the<br>op-point) → the transfer $H_{\{(k1,k2)\}} =$<br>$\sum_j \Psi_{\{(k1,k2), j\}} \cdot B0_j$ from an input at<br>every sideband $f_{\text{in}} + k1 \cdot f1 + k2 \cdot f2$ to the<br>output, all from ONE adjoint. By the<br>reciprocity identity<br>$(H^{-1}B)_{\text{out}} = (H^{-T}e_{\text{out}})^T B$ the<br>sideband- $(0,0)$ transfer is bit-identical to the<br>QPAC response (the PXF↔PAC cross-check,<br>E-125). Reads only retained data →<br>solver-independent (KLU + Sparse). Verified<br>6/6: reciprocity (QPXF(0,0)==QPAC(0,0)<br>bit-identical), conversion-sideband<br>reciprocity, reduce-to-XF (pump→0 ?<br>$(0,0)$ =plain transfer=R, sidebands vanish),<br>clean no-op-point error, KLU==Sparse;<br>QPSS 11/11 + QPSS-HB 7/7 + QPAC 7/7 +<br>QPnoise 10/10 unaffected |

| Enhancement | ngspice files                                                                                                                                                                   | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-142       | spicelib/analysis/dcpss.c,<br>frontend/com_qpac.{c,h}, com_qpnoise.c,<br>com_qpxf.c, commands.c,<br>include/ngspice/cktdefs.h, exam-<br>ples/qpss_examples/verify_qpss_sweep.py | input-frequency sweep for the two-tone<br>small-signal analyses. qpac/qpnoise/qpxf<br>gain a <dec oct lin> N fstart fstop<br>sweep of f_in that emits a plottable ngspice<br>plot (conversion gain / noise figure /<br>image-rejection curves vs frequency),<br>matching how .ac/.pnoise/.pxf sweep.<br>QPACsweep/QPnoiseSweep/QPXFstep<br>f_in and reuse the exact single-frequency<br>solve per point: qpac records per-node<br> (0,0) response , qpnoise records<br>onoise_spectrum/inoise_spectrum, qpxf<br>records xf (in-band (0,0)) + xf_conv (total<br>conversion $\sqrt{\sum  H_{\{sb \neq 0\}} ^2}$ ). The analysis<br>fills plain arrays; the front-end builds the plot<br>with a frequency scale via the nutmeg vector<br>API (plot_alloc/dvec_alloc/vec_new) —<br>the correct layer for a front-end command<br>(the analysis OUTp framework dereferences<br>a live analysis job's JOBtype, which a<br>front-end command lacks → was crashing).<br>Shared helpers<br>qp_steptype/qp_sweep_maxpts/qp_emit_plot.<br>Verified 5/5: swept value at 0.3G ==<br>single-frequency qpac/qpxf/qpnoise<br>(machine precision), reactive roll-off vs f_in,<br>correct point count; single-frequency QPAC<br>7/7 + QPnoise 10/10 + QPXF 6/6 unaffected |

| Enhancement | ngspice files                                                                                                                                               | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-143       | frontend/com_optimize.c,<br>frontend/commands.c, exam-<br>ples/optimize_examples/verify_optimize.py,<br>exam-<br>ples/optimize_examples/optimize_lsq_demo.c | gradient least-squares curve fitting for <b>optimize</b> . The built-in optimizer (E-130, derivative-free Nelder-Mead on a scalar <code>-minimize</code> ) gains a least-squares mode: one or more weighted <code>-target &lt;expr&gt; &lt;value&gt; [&lt;weight&gt;]</code> measurements, optionally spread over several <code>-analysis</code> stages (each <code>-analysis</code> opens a stage; following <code>-targets</code> attach to it, so a single fit can combine e.g. a DC operating point AND an AC response), are fitted with Levenberg-Marquardt — a forward/backward finite-difference Jacobian $J$ , normal equations $A=J^T J$ , $g=J^T r$ , damped solve $(A+\lambda \cdot \text{diag}(A)) \delta = -g$ via a small partial-pivot Gauss solver, standard $\lambda$ up/down loop. Exploiting the sum-of-squares structure it converges in far fewer analysis runs than the simplex (27 vs 67 evals on a two-target RC fit). <code>-method nm lm</code> overrides the auto choice (LS $\rightarrow$ lm, scalar $\rightarrow$ nm); the scalar <code>-minimize</code> /Nelder-Mead path is unchanged; <code>-minimize</code> and <code>-target</code> are mutually exclusive. Parameters are still applied in place with <code>alter</code> (no re-source) and the search runs in normalized [0,1] space. All changes in <code>com_optimize.c</code> (+ one help string); no new files, no ABI change. Verified 23/23 (E-130 checks [1]–[5] + E-143 [6]–[10]): 1-param/3-target/3-stage RC magnitude fit, 2-param DC+AC multi-analysis fit ( $R1=3k, R2=2k$ ), LM-fewer-evals-than-NM, OSDI/Verilog-A diode extraction (recover both $is \rightarrow 1e-14$ and $n \rightarrow 1.2$ from two measured I-V points by weighted least squares), and input validation (lm-without-target, minimize+target, multi-analysis+scalar all rejected) |

| Enhancement | Verilog files                                                                                                                                                | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-144       | frontend/com_optimize.c, frontend/inp.c, frontend/commands.c, examples/optimize_examples/verify_optimize.py, examples/optimize_examples/optimize_dparam_demo | <p><b>optimize</b> tunes symbolic <b>.param</b> values via a new knob kind <b>-dparam</b>. <b>-param</b> remains an alter target (device/instance, changed in place); <b>-dparam</b> names a netlist <b>.param</b> symbol (e.g. <b>.param w=1u, R1={500*k}</b>) which — being expanded at parse time — is changed with <b>alterparam &lt;name&gt;=&lt;value&gt;</b> + a reset that re-sources the deck (re-evaluating <b>.param</b> expressions and re-stamping device values). Per parameter a kind (<b>OPT_ALTER</b> / <b>OPT_DECKPARAM</b>) is tracked; when any <b>-dparam</b> is present <b>opt_eval</b> applies ALL deck params first, re-sources ONCE, THEN applies the in-place alter params (reset rebuilds from the deck and would otherwise wipe an earlier alter — this ordering is what lets the two kinds mix; verified on a joint <b>.param</b>+device fit). No <b>-dparam</b> the E-130/-143 fast path is untouched (no reset, no perf hit). The two re-source banners (<b>Reset re-loads circuit ...</b> and <b>Circuit: ...</b> in <b>inp.c</b>) are gated by the existing <b>ft_optimizing</b> flag so hundreds of inner re-sources are silent (only the final leave-at-optimum run shows); <b>ft_optimizing</b> re-asserted after each reset. Also fixed 5 pre-existing <b>-Wsign-conversion</b> warnings in <b>inp.c</b>'s <b>*ng_script_with_params</b> handler (<b>argc/size</b> unsigned). Closes the last E-143 follow-up. No new files, no ABI change. Verified 31/31 (E-130/-143 [1]–[10] + E-144 [11]–[15]): scalar <b>.param</b> fit (<b>rtop</b>→2333.3), mixed <b>-dparam</b>+<b>-param</b> joint fit (<b>rtop</b>=3k,<b>R2</b>=2k), <b>.param</b> in expression (<b>k</b>→6), least-squares <b>-dparam</b> (LM), quiet re-source (<math>\leq 1</math> banner); AC + OSDI-device-value <b>.param</b> verified manually</p> |

| Enhancement | mspice files                                                                                                                                                                           | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-145       | frontend/com_optimize.c, frontend/commands.c, examples/optimize_examples/verify_optimize.py, examples/optimize_examples/optres.m.va, exam- ples/optimize_examples/optimize_mparam_demo | <b>optimize</b> tunes <b>.model</b> -card parameters via a third knob kind <b>-mparam</b> . <b>-param</b> remains an alter instance target and <b>-dparam</b> a symbolic <b>.param</b> (re-source); <b>-mparam</b> <b>&lt;@model[param]&gt;</b> names a <b>.model</b> -card parameter (e.g. <b>@dmod[is]</b> , <b>@rmod[r]</b> ) — which is NOT alter-reachable and NOT <b>.dc</b> -sweepable (only sources/resistors/instance params are) — and is changed in place with <b>altermod</b> <b>&lt;name&gt;=&lt;value&gt;</b> , no re-source (as cheap per eval as <b>-param</b> , unlike <b>-dparam</b> ). A new kind value <b>OPT_MODELPARAM</b> ; in <b>opt_eval</b> the deck params are re-sourced first (E-144), then the in-place knobs: <b>alter</b> for instances, <b>altermod</b> for models. <b>@model[param]</b> is passed straight to <b>altermod</b> (which accepts that accessor form), mirroring how <b>-param</b> <b>@m1[w]</b> emits <b>alter @m1[w]=...</b> . Circuits with only <b>-param/-mparam</b> keep the E-130/-143 fast path (no reset). All three knob kinds mix in one run. Change confined to <b>com_optimize.c</b> (+1 help line); no new source files, no <b>inp.c</b> change, no ABI change. Verified 39/39 (E-130/-143/-144 [1]–[15] + E-145 [16]–[20]): OSDI model-param fit ( <b>@rmod[r]</b> → 3k), built-in diode model-param fit ( <b>@dmod[is]</b> → 1.22e-14), determined model+instance joint fit (r=3k, R2=2k), <b>-mparam</b> does 0 re-sources (in-place fast path), and all three kinds ( <b>-dparam+-mparam+-param</b> ) coexist and converge |

| Enhancement | ngspice files                                                                                                                                                                        | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-146       | frontend/com_sweep.c (new),<br>frontend/com_sweep.h (new),<br>frontend/commands.c,<br>frontend/com_commands.h, frontend/inp.c,<br>frontend/Makefile.am,<br>examples/sweep_examples/* | universal <b>sweep</b> command + <b>.sweep</b> card. Sweeps ANY circuit knob, generalizing <b>.dc</b> (which steps only sources/resistors/instance params). <b>sw_kind</b> auto-detects the knob: <b>@X[y]</b> with <b>ft_sim-&gt;findModel(ckt,X)</b> non-NULL <b>?</b> model param ( <b>altermod</b> ), else instance ( <b>alter</b> ); a bare name with <b>nupa_get_param(name,&amp;found)</b> found <b>?</b> <b>.param</b> ( <b>alterparam+reset</b> ), else device ( <b>alter</b> ) — so model params and <b>.params</b> , impossible with <b>.dc</b> , are now sweepable. <b>sweep &lt;knob&gt; (&lt;start&gt; &lt;stop&gt; &lt;step&gt;   lin dec oct N a b   list ...)</b> <b>[-analysis &lt;cmd&gt;] [-output &lt;name&gt;=&lt;expr&gt; ...]</b> : sets the knob, runs the inner analysis (default op) at each point, evaluates each output (last value; default = all node voltages like <b>.dc</b> ), and emits a summary plot named <b>sweep</b> with the knob values as the scale (nutmeg vector API, front-end layer per E-142). The per-point analysis plots are retained ( <b>tran1,tran2,...</b> ). <b>.sweep</b> card: <b>inp.c</b> collects a top-level <b>.sweep</b> line into the post-parse control list as a <b>sweep</b> command; a static <b>sweep_active</b> re-entrancy guard stops a <b>.param</b> re-source ( <b>reset</b> re-runs the <b>.sweep</b> card) from recursing. New source files registered in <b>commands.c/com_commands.h/Makefile.am</b> . Verified 11/11 ( <b>verify_sweep.py</b> ): <b>sweep R1</b> reproduces built-in <b>.dc R1</b> bit-for-bit, model-param ( <b>@rmod[r]</b> ) + <b>.param(rtop)</b> sweeps vs analytic divider, auto-detection routing, <b>.sweep</b> card == command form (no recursion), AC named-output vs analytic <b> H(1kHz) </b> , transient settled value, <b>lin/list/step</b> point counts, multi-output |

| Enhancement | ngspice files                                                                                                                                                             | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-149       | maths/misc/randnumb.c,<br>include/ngspice/randnumb.h,<br>frontend/numparam/xpressn.c,<br>frontend/numparam/spicenum.c,<br>frontend/commands.c,<br>examples/lhs_examples/* | <p>Latin-Hypercube Monte Carlo sampling (mcsample), closing the low-discrepancy-sampling gap vs commercial tools. A stratified sampler in randnumb.c holds the mode/N/sample-index/dimension-counter/seed; each random dimension's stratum permutation (Fisher-Yates over <math>0..N-1</math>) + per-sample jitter are generated lazily from a self-contained splitmix64 seeded by (seed, dimension) — reproducible and order-independent.</p> <p>mc_sample_uniform() = (perm[d][i]+jitter[d][i])/N;<br/> mc_sample_gauss() maps it through inv_normal_cdf (Acklam probit, rel err &lt; 1.15e-9) so Gaussian tails stratify too. Sample boundary: spicenum.c's nupa_signal(NUPADECKCOPY) edge (once per reset re-source, guarded by the existing firstsignals latch) calls mc_sample_advance() (step sample, rewind dimension) before that pass's .param draws. Draw hook (xpressn.c): agauss/gauss take one mc_sample_gauss() and aunif/unif/limit one <math>2 \cdot mc\_sample\_uniform() - 1</math> when LHS is active, else the unchanged gauss1()/drand() (LHS-off is byte-identical to before). Command com_mcsample (randnumb.c, registered in commands.c both tables) parses lhs &lt;N&gt; [seed &lt;s&gt;] / random / off. Front-end only, so solver-independent. Verified 5/5 under both solvers (verify_lhs.py): stratification (each of 48 strata hit once vs random's ~32/48), two-param multi-dimension, ~130x lower Var(sample-mean) at N=32 (both converge to the same mean), seed reproducibility, analytic mean/sigma correctness</p> |

| Enhancement | ngspice files                                                                                                                                                  | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-150       | frontend/com_sweep.c,<br>frontend/commands.c,<br>maths/misc/randnumb.c,<br>include/ngspice/randnumb.h,<br>examples/_setup.py,<br>examples/highsigma_examples/* | <p>high-sigma rare-event estimation (highsigma), the last statistical ROW versus a commercial simulator -- 4-6 sigma failure probabilities (SRAM / standard-cell yield) that plain MC cannot reach (1e-7..1e-9 needs 1e7..1e9 runs). Scaled-sigma importance sampling: a new MC_MODE_SSS in the E-149 sampler draws each Gaussian .param from the lambda-inflated normal (<math>z = \text{lambda} * \text{gauss1}()</math>) and accumulates that draw's log likelihood-ratio <math>\log(\text{lambda}) - (z^2/2)(1 - 1/\text{lambda}^2)</math> into sss_logw; <math>\text{mc\_sample\_weight}() = \exp(\text{sss\_logw})</math> reweights the sample so the estimator is unbiased for the nominal probability. Uniform .params are bounded so SSS leaves them un-inflated (weight 1). Direction-free -- no gradient / sensitivity / most-probable-failure-point.</p> <p><code>mc_sss_config(N, lambda, seed)</code> seeds the global PRNG for reproducibility; <code>mc_sss_off()</code> reverts. Command <code>com_highsigma</code> lives in <code>com_sweep.c</code> (reuses its <code>sw_run_cmd</code> reset/analysis runner and <code>sw_eval_expr</code>), registered in <code>commands.c</code>:</p> <p><code>highsigma &lt;N&gt; [-scale &lt;lambda&gt;] [-seed &lt;s&gt;] [-analysis &lt;cmd&gt;] [-metric &lt;expr&gt; [-max &lt;hi&gt;] [-min &lt;lo&gt;]</code> loops N samples (reset -&gt; analysis -&gt; eval metric -&gt; spec test -&gt; read weight), reports P(fail), relative error, equivalent one-sided sigma-to-fail (<math>-\Phi^{-1}(P)</math>) and failure count, and leaves them in the <code>highsigma_*</code> vectors/vars. The spec is <code>-max/-min</code> values (not a <code>&gt;/&lt;</code> inside the expr, which a control command reads as an I/O redirect). Front-end only, solver-independent; the verify is heavy (thousands of re-sources) so <code>examples/_setup.py</code> marks highsigma <code>SPARSE_ONLY</code>. Verified vs analytic <math>\Phi(-\beta)</math> (<code>verify_highsigma.py</code>): <math>\beta=4 \rightarrow</math> sigma 4.000, P 3.16e-5 (true 3.17e-5); deep tail <math>\beta=5</math> (P 2.87e-7) recovered from 6000 runs where plain MC expects ~0.0017 failures; two-sided spec ~doubles P; seed reproducibility exact; two independent Gaussians combine as <math>N(., \sqrt{s_1^2 + s_2^2})</math></p> |

| Enhancement | ngspice files                                                                                                                                                                                                                                                                                      | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-151       | maths/misc/randnumb.c,<br>include/ngspice/randnumb.h,<br>frontend/numparam/xpressn.c,<br>frontend/com_sweep.c,<br>frontend/commands.c, examples/_setup.py,<br>examples/yield_examples/*                                                                                                            | process/mismatch correlations + packaged Monte Carlo yield -- closes the last two partial (warn) statistical rows vs a commercial simulator. (1) CORRELATIONS: <code>mc_corr_config(k, matrix)</code> Cholesky-factors a $k \times k$ correlation matrix (row-major, unit diagonal; rejects a non-positive-definite one) into lower-triangular <code>corr_L</code> ; <code>mc_corr_component(i)</code> lazily draws the $k$ underlying $z$ 's once per sample THROUGH <code>mc_sample_gauss()</code> (so LHS stratification / SSS weighting apply), computes $y = L*z$ , caches it, and returns $y[i]$ . The cache is reset in <code>mc_sample_advance()</code> which now runs its reset in EVERY mode (so correlation works under plain MC, not just LHS/SSS). New <code>.param</code> function <code>mvnorm(i)</code> (frontend/numparam/xpressn.c: added to <code>fmathS</code> , <code>XFU_MVNORM</code> enum in <code>fmathS</code> order, and the dispatch) returns the $i$ -th correlated standard normal; with no matrix registered it degrades to an independent draw. <code>com_mccorr(mccorr &lt;k&gt; &lt;m11&gt; . . &lt;mkk&gt;</code> KLU matrix reordering + scaling controls. The KLU direct solver ran on the compiled-in <code>klu_defaults</code> (AMD ordering, max row scaling, BTF on) with no way to change them, and its one exposed knob <code>klu_memgrow_factor</code> was broken ( <code>task-&gt;TSKkluMemGrowFactor = (val-&gt;rValue == 1.2)</code> stored a boolean, not the value). E-152 exposes three new <code>.options</code> -- <code>'klu_ordering=amd</code> Levenberg-Marquardt trust-region Newton ( <code>.option trustregion</code> , off by default), completing the damped/trust-region-Newton row alongside the E-111 line search. In the Newton loop ( <code>niiter.c</code> ): the E-111 KCL-residual merit is reused (gated on <code>linesearch</code> |
| E-152       | include/ngspice/optdefs.h,<br>include/ngspice/tskdefs.h,<br>include/ngspice/cktdefs.h,<br>include/ngspice/smpdefs.h,<br>spicelib/analysis/cktsopt.c,<br>spicelib/analysis/cktntask.c,<br>spicelib/analysis/cktdjob.c,<br>maths/ni/niinit.c, maths/KLU/klusmp.c,<br>examples/klu_tuning_examples/*  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| E-153       | maths/ni/niiter.c, maths/KLU/klusmp.c,<br>include/ngspice/smpdefs.h,<br>include/ngspice/optdefs.h,<br>include/ngspice/tskdefs.h,<br>include/ngspice/cktdefs.h,<br>spicelib/analysis/cktsopt.c,<br>spicelib/analysis/cktntask.c,<br>spicelib/analysis/cktdjob.c,<br>examples/trustregion_examples/* |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| E-154 | frontend/com\_envelope.c (new), frontend/com\_envelope.h (new), frontend/commands.c, frontend/com\_commands.h, frontend/Makefile.am, spicelib/analysis/envelope.c (new), spicelib/analysis/Makefile.am, include/ngspice/cktdefs.h | Envelope Following (`envelope <node> <fc> <tstop> [nppp] [m] [maxm] [reltol] [set` the last missing analysis in the RF/periodic-steady-state suite. For a carrier-driven circuit whose amplitude/phase envelope varies slowly over many carrier periods, it samples the state once per carrier period  $T=1/f_c$  and integrates the slow drift, jumping  $M$  periods at a time. The exact per-period map

is  $X_{n+1} = \text{phi}(X_n)$ ,  $\text{phi}$  = one-period DAE integration; the envelope obeys  $dX/dn \sim \text{phi}(X) - X$ . The naive forward-Euler jump  $X_{n+M} = X_n + M(\text{phi}(X_n) - X_n)$  is UNSTABLE on high-Q circuits (the one-period monodromy has unit-circle eigenvalues, so  $I + M(\text{Phi} - I)$  amplifies  $\rightarrow$  blow-up; this is what shelved an earlier attempt). The new engine (spicelib/analysis/envelope.c, EFanalysis) uses the IMPLICIT backward-Euler jump  $X_{n+M} = X_n + M(\text{phi}(X_{n+M}) - X_{n+M})$ , solved by Newton  $[(1+M)I - M*\text{Phi}] dY = -G$  with  $\text{Phi} = d\text{phi}/dY$  the one-period monodromy (finite-differenced, one extra period-integration per state, dense NxN solve capped for modest circuits). A-stable, so it tracks a resonator's envelope without diverging; its fixed point  $\text{phi}(X) = X$  is the true steady state. Step size  $M$  is chosen by step-doubling local-truncation-error control. The one-period map reuses the transient primitives (NicomCof + Nlitter per fixed sub-step, the dctrans state rotation) on a fixed grid of nppp points in TRAPEZOIDAL mode (backward-Euler damps a high-Q resonance);  $\text{phi}$  is SELF-STARTING (a frozen-history variant plateaued at a false low fixed point) with a sub-divided backward-Euler first sub-step to bound the restart damping. The com\_envelope.c front-end command runs a short settling transient, calls EFanalysis, and emits an envelope plot ( $\text{\_amp} = 2|V_1|$ ,  $\text{\_dc}$ ,  $\text{\_re}$ ,  $\text{\_im}$  vs time; nutmeg vector API). Solver-independent. Verified against a full .tran under both linear solvers: EF amplitude tracks the transient across a Q~3160 tank ring-up to <3% and to the steady state (<0.5%), stays bounded over 3000 carrier periods (26 envelope samples), and tracks a Q~316 tank to ~1.6%. Completes the RF suite |

| E-155 | frontend/com\_reduce.c (new), frontend/com\_reduce.h (new), frontend/commands.c, frontend/com\_commands.h, frontend/Makefile.am, spicelib/analysis/rcreduce.c (new), spicelib/analysis/Makefile.am, include/ngspice/cktdefs.h, examples/reduce\_examples/\* | RC network reduction (reduce <fmax> [factor f] [file fname] [name subckt] [keep node ...]), moving the post-layout "RC reduction / model-order reduction" gap to done. Post-layout extraction produces enormous linear parasitic R/C networks (thousands-millions of interior nodes); reduce collapses the R/C network to a small ELECTRICALLY EQUIVALENT one preserving the port behaviour over DC..fmax, written as an ordinary .subckt of R's and C's. Method (spicelib/analysis/rcreduce.c, CKTreduceRC): TICER (Time-Constant Equilibration Reduction) -- Schur-complement (Gaussian) elimination of interior nodes of  $Y(s) = G + sC$ , kept first order in  $s$  so the reduced network is REALIZABLE as R's and C's (no model-order-reduction black box, no passive-synthesis step). Per elimination of node  $n$ , for each neighbour pair  $(a,b)$ :  $G[a,b] := G[a,n]G[n,b]/G[n,n]$ ;  $C[a,b] := (G[a,n]C[n,b] + C[a,n]G[n,b])/G[n,n] - G[a,n]G[n,b]C[n,n]/G[n,n]^2$ . The conductance update is the exact Schur complement so DC is preserved EXACTLY; the capacitance is matched to first order. A node is eliminated only when its self time-constant frequency  $f_n = G_n/(2\pi C_n) > \text{factor} * f_{\text{max}}$  (quasi-static in the band of interest); factor trades reduction against in-band accuracy, monotonically. PORTS (kept nodes) are auto-detected as every node touched by a device that is NOT a resistor/capacitor (sources, transistors, OSDI Verilog-A devices), plus ground and user keep nodes, via the generic GENnode/terminal-count device interface -- so built-in and OSDI devices alike. The com\_reduce.c front-end command enumerates the built-in resistor/capacitor instances (CKTtypelook), builds dense G/C over the RC nodes, runs TICER, and emits the reduced subckt (CKTnodName for node names). Solver-independent (reads devices + own dense solve). First cut is dense (node cap ~2500); sparse TICER is the follow-up. Verified under both linear solvers: a huge factor eliminates nothing and the emitted subckt reproduces the full AC response BIT-for-BIT; a moderate factor cuts nodes (e.g. 25  $\rightarrow$  6, 4x) with <0.25 dB in-band error and DC exact; the accuracy/reduction tradeoff is monotone in factor; an OSDI device on a node auto-marks it a kept port |

| E-156 | spicelib/analysis/rcreduce.c, frontend/com\_reduce.c, include/ngspice/cktdefs.h, examples/reduce\_examples/\* | Sparse RC reduction -- makes the E-155 reduce command scale to real post-layout size. The dense NN TICER engine (capped ~2500 nodes) is replaced by a SPARSE one: the network is stored as per-node adjacency lists (RCadj: a growable edge list of {neighbour, g, c}), and interior nodes are eliminated in a MINIMUM-DEGREE order (a lazy binary heap keyed on current degree, like sparse LU) so fill-in stays tiny -- a degree-2 chain node merges two series elements with ZERO fill. A FILL GUARD (maxdeg, default 12) declines to eliminate a node once its degree grows past the threshold, so a dense mesh core (whose boundary Schur complement is inherently dense) is left intact instead of densifying

the whole network (measured: unguarded, one mesh elimination created 3308 fill edges; guarded, 9). The TICER Schur-complement update, DC-exact conductance, frequency criterion (eliminate only when  $f_n = G_n / (2\pi C_n) > \text{factorfmax}$ ), and automatic port detection are unchanged from E-155; in the edge representation the diagonal is implicit so eliminating a node just adds Schur edges among its neighbours. Node cap raised 2500 -> 5,000,000; a 65,017-node network reduces in ~4 s. New maxdeg command option (com\_reduce.c) and CKTreduceRC parameter (cktdefs.h). Also fixes a terminal-ordering pitfall: the reduced .subckt's terminals are emitted in internal-index order (not the order keep nodes were typed), so instantiating with the wrong order silently swaps ports -- the command now prints the correct x1 <ports...> <name> instantiation line. Verified under both linear solvers: identity reduction bit-exact (0.00 dB), a moderate factor gives ~4x fewer nodes at <0.25 dB in-band with DC exact, monotone accuracy/reduction tradeoff, OSDI auto-port, and an 8001-node network (past the old dense cap) reduces successfully | | E-157 | frontend/com\_aging.c (new), frontend/com\_aging.h (new), frontend/commands.c, frontend/com\_commands.h, frontend/Makefile.am, examples/aging\_examples/\* | Device aging (aging <t\_target> [rate <opvar>] [param <ageparam>] [dynamic <tstop> [tstep]] [verbose]), adding the reliability "stress -> degrade -> re-simulate (fresh/aged)" flow (HCI/NBTI/TDDDB) and moving those reliability gap rows to done. The command finds every aging-capable device in the loaded circuit, computes how much it has degraded after t\_target seconds of operation, writes that back into the device, and RE-STAMPS the circuit so any analysis run afterwards (op/dc/tran/ac) sees the aged devices; it runs the fresh stress simulation first, leaving it as the current plot (a "fresh" baseline). The design is MODEL-AGNOSTIC: a device opts in purely by exposing, in its Verilog-A/OSDI model, a degradation-RATE operating-point variable (default agerate, in dose-units/second) and a per-instance AGE parameter (default age, (\*type="instance"\*)). The engine's only job is to integrate the rate into a dose and feed it back; the model owns the physics that maps age to a parameter shift (a sublinear power law, an Arrhenius factor, ...). Participants are detected by scanning each device TYPE's instance-parameter table (DEVices[t]->DEVpublic.instanceParms) for the two keywords, so ordinary resistors/sources are silently skipped and probing never errors. Two modes: STATIC (default) reads the rate at the DC operating point and scales by the lifetime, age = agerate(op)t\_target; DYNAMIC (dynamic <tstop>) runs a transient over one representative window, trapezoidally (time-weighted) integrates the rate, and extrapolates, age = (INT agerate dt / tstop)t\_target -- capturing duty cycle. Because age is per-instance, devices sharing a .model but at different bias age independently. Implementation (frontend/com\_aging.c) reuses the com\_optimize synchronous command-dispatch pattern to run op/tran, the nutmeg expression engine to read @inst[agerate], and alter @inst[age]=... to write back; console chatter suppressed via ft\_optimizing unless verbose. Solver-independent. Verified under both linear solvers with a demo square-law NMOS carrying an NBTI hook (agemos.va): enumeration picks exactly the ageable devices, the dose is exactly ratet\_target, the threshold shift matches the analytic  $\Delta V_{th} = dv_{th\_ref}(\text{age}/\text{age\_ref})^{0.25}$  law to 5 sig figs, aged drain current drops, degradation is monotone in lifetime, a near-threshold device loses a larger fraction of its current than a hard-driven one for the same shift, a 30%-duty pulsed gate ages at 0.30x the DC rate (dynamic), and a sub-threshold device accrues zero dose | | E-158 | frontend/com\_emir.c (new), frontend/com\_emir.h (new), frontend/commands.c, frontend/com\_commands.h, frontend/Makefile.am, examples/emir\_examples/\* | Power-grid EMIR (emir [rail V] [thresh frac] [thick m] [jmax A/m2] [n exp] [tref s] [top k] [verbose]), adding electromigration + IR-drop reliability sign-off and moving the last reliability gap row to done. After a DC solve of the power-distribution network the command reports two metrics. IR-DROP: for every node, the drop below the ideal rail (rail - V(node)); the rail defaults to the highest node voltage (the supply pad) unless given; reports the worst node and every node past a threshold (default 10% of the rail). ELECTROMIGRATION: for every wire-segment resistor, the current DENSITY  $J = |I|/(w\text{thickness})$ , ranked; the worst-density segment, every segment past the current-density limit Jmax, and a Black's-equation relative lifetime MTTF/ref = (Jmax/J)^n (Black: MTTF ~ J^-n; a segment exactly at Jmax has MTTF = the reference lifetime tref, needing no process-specific prefactor). The physics the analysis exists to expose: EM is set by current DENSITY, not current -- a fat trunk carrying huge current can be safe while a thin wire at a fraction of that current voids first, which is why real grids taper wire width with carried current to hold J roughly constant. Implementation (frontend/com\_emir.c) runs a fresh op (com\_optimize synchronous-dispatch

pattern), walks the current plot's node-voltage vectors (`plot_cur->pl_dvecs` filtered to `SV_VOLTAGE`) for IR-drop, and enumerates the resistor instances (the device type named "Resistor") reading each segment's current and width via the nutmeg expression engine (`@Rk[i]`, `@Rk[w]`) -- so it needs no device-struct access and works for any resistor; segments without a width are skipped and counted. Both tables are qsort-ranked (IR by drop, EM by density) and truncated to top. Solver-independent (reads a DC solution + per-resistor currents). Verified under both linear solvers on a 3-segment tapered ladder off a 1 V rail: worst IR drop 0.30 V (30%) at the far tap matching the exact ladder solve and linear in load current;  $J = I/(w\text{thick})$  exact; the worst-density segment is the narrow low-current wire, not the high-current wide trunk; the Black MTTF ratio between two segments equals  $(J2/J1)^n$ ; with  $J_{\max}$  between the trunk and leaf density exactly the under-sized segments fail; rail auto-detect equals an explicit rail; and an OSDI (Verilog-A) current load at a tap is handled identically to a current source | | [E-159](#) | (no ngspice source change) `examples/compactmodels_examples/*` | Real production compact-model bring-up -- a validation/example enhancement that exercises the EXISTING toolchain on actual CMC (Compact Model Coalition) Verilog-A models, so no ngspice or openvaf-r source change. BSIM4 (the ~12,600-line industry-standard bulk MOSFET) is compiled through openvaf-r to OSDI (~6 s) and validated against ngspice's BUILT-IN BSIM4 -- the same model, so a rigorous self-check: the OSDI BSIM4.8 and native BSIM4.8.3 agree to ~2% on the Id-Vds output family and ~4.5% on the Id-Vgs transfer curve (the residual is the point-release version gap). EKV (EPFL compact MOSFET, which ngspice has NO built-in for) is brought up purely through OSDI, giving textbook saturation. A general finding surfaced: OSDI keeps every internal node STATIC (no dynamic node collapsing), so BSIM4's internal drain/source nodes (`di/si`) -- which the default `rdsmod=0` expects the simulator to collapse onto the external d/s -- are left floating and the device conducts ZERO current; enabling the external series-resistance nodes with `rdsmod=1` (the model near-shorts them, 1e3 mho, when no S/D geometry is given) connects them and the device works. Models are compiled in place from the bundled OpenVAF integration-test sources (licenses stay with them). Verified under both linear solvers (6 checks) | | [E-160](#) | (no ngspice source change) `examples/cmcsweep_examples/`, `examples/_setup.py` | CMC compact-model coverage sweep -- the comprehensive follow-up to E-159, again a validation/example enhancement with no ngspice/openvaf-r source change (only `_setup.py` gains `cmcsweep` in its `SPARSE_ONLY` and `REGRESSION_EXCLUDE` sets so the slow 19-model compile runs single-solver and off the fast sweep). It drives EVERY real CMC (Compact Model Coalition) Verilog-A model bundled with OpenVAF through openvaf-r -> OSDI -> ngspice and reports a coverage matrix, the same idea as the E-84 LRM sweep but for production compact models. Result: 19 of 19 models COMPILE and 19 of 19 LOAD, spanning every device class -- MOSFET (BSIM3, BSIM4, BSIM6, BSIMBULK, EKV, HiSIM2, HiSIMSOTB, MVSG\_CMC), FinFET (BSIM-CMG, BSIMIMG), SOI (BSIMSOI, HiSIMHV), GaN HEMT (ASMHEMT), surface-potential (PSP102, PSP103), bipolar/SiGe-HBT (HICUML2, MEXTRAM), and diode (DIODE, DIODE\_CMC). Deeper conduction+physics validation for a representative model per class: OSDI BSIM4 and BSIM3 match ngspice's BUILT-IN BSIM4/BSIM3 to ~2%/~7%; EKV gives monotonic saturating I-V; HICUML2 shows bipolar current gain  $\beta \sim 100$  on a Gummel plot; DIODE\_CMC turns on exponentially ( $\times 16000$  over 0.2-1.0 V). Two findings: (1) the E-159 internal-node/`rdsmod=1` issue is BSIM4-SPECIFIC, not universal -- BSIM3 handles the zero-resistance case fine and conducts out of the box (its apparent zero at  $V_{gs}=1$  V is just a high ~1.7 V default threshold); (2) HiSIMHV (6 terminals `d,g,s,b,sub,temp`) needs `cosubnode=1` to enable the substrate node, and with the default `cosubnode=0` its `$fatal` correctly guards the node-count mismatch (confirming `$fatal` works through OSDI). Models compiled in place from the bundled integration-test sources (licenses stay put). Verified (7 checks); the compile/load tiers are solver-independent | | [E-161](#) | (no ngspice source change) `examples/dynmodels_examples/` | Dynamic (AC/RF) compact-model validation -- extends the E-159/160 DC validation into the dynamic domain, again a validation/example enhancement with no ngspice/openvaf-r source change. Where E-159/160 exercised DC I-V, this exercises the DYNAMIC behavior, which flows through a different code path: OSDI's REACTIVE (charge) Jacobian stamping and ngspice's `.ac` analysis, on the real production models (compiled in place from the bundled CMC sources). BSIM4 C-V: the gate capacitance  $C_{gg}(V_{gs})$  is extracted from `.ac` as  $\text{Im}(I_{\text{gate}})/\omega$  and swept over gate bias; it rises from ~40 fF in subthreshold (overlap+fringe) to ~129 fF in inversion (approaching  $C_{ox}WL$ ) -- the textbook MOSFET C-V -- and the OSDI-compiled BSIM4 matches ngspice's BUILT-IN BSIM4 to under 1% at every bias (a tighter check than the ~2% DC

I-V match, since Cgg is dominated by the version-independent oxide term). Cutoff frequency  $f_T$  (the frequency where AC current gain  $|h_{21}|=|I_{out}/I_{in}|$  falls to 1): OSDI BSIM4  $f_T \sim 3.5$  GHz matches the built-in to  $\sim 1\%$ ; HICUML2 (SiGe HBT, which ngspice has no built-in for) has zero default transit time ( $t_0=0$ ) giving infinite  $f_T$ , so a realistic dynamic parameter set ( $t_0=10$  ps, 1 fF junction caps) is supplied and the resulting  $f_T$  lands right at the transit-time limit  $1/(2\pi t_0) \sim 15.9$  GHz and rises with collector current -- textbook charge-control behavior. Verified under BOTH the Sparse and KLU solvers (.ac is supported by both), 4 checks | | [E-162](#) | frontend/inp.c, examples/hb\_examples/verify\_hb.py | **.hb** dot-card for harmonic balance -- gives the E-134 hb command netlist dot-card parity with the PSS family (.pss/.pac/.pnoise/.pxf/.psp/.sp are dot-cards, while the HB family hb/qps/hbosc were control-block commands only). A top-level **.hb** <f0> <K> [points] [maxiter] card now runs single-tone harmonic balance straight from the deck (no .control block needed). Rather than duplicating the HB machinery as a separate analysis JOB, **.hb** reuses the deck->control mechanism Enhancement-146 introduced for .sweep: during deck loading (frontend/inp.c) a top-level **.hb** ... line is stripped of its leading . and appended to the post-parse control list, so it executes as hb ... (the com\_hb command) once the circuit is built -- inheriting all of the E-134 engine (argument parsing, the E-121 conversion-matrix Jacobian, E-135 source-stepping continuation, solver independence). A boundary check (next char after .hb is whitespace or end-of-line) keeps it from swallowing any future **.hb\*** card. Before this, **.hb** produced unimplemented dot command '**.hb**' -- the only **.hb** handler in ngspice was a disabled `#ifdef WITH_HB` upstream stub (dot\_hb) that merely redirected to the PSS analysis, not the E-134 engine. Verified (examples/hb\_examples/verify\_hb.py, +2 checks): the **.hb** dot-card run in plain batch mode (no .control) converges and produces a harmonic-balance spectrum bit-for-bit identical to the hb command form on a junction-limited diode rectifier; it threads its optional [points]/[maxiter] arguments and coexists with a following .control block (deck order preserved); both Sparse and KLU give identical results. (Like .sweep, a bare command-style dot-card with no other analysis card prints a benign "no simulations run" batch notice even though HB ran.) | | [E-163](#) | frontend/inp.c, examples/qps\_examples/verify\_qps.py, examples/phasenoise\_examples/verify\_phasenoise.py | **.qps** / **.hbosc** / **.phasenoise** dot-cards -- completes the harmonic-balance family's netlist dot-card parity begun by E-162's **.hb**. Three more top-level command-style cards now run their analyses straight from the deck: **.qps** <expr> <f1> <f2> [periods] [maxorder] (two-tone quasi-periodic steady state, E-133/136; add the hb keyword for the frequency-domain form), **.hbosc** <oscnode> <K> [fguess] [tstab] (autonomous harmonic balance for an oscillator, E-140; the deck needs a .ic to start the oscillation), and **.phasenoise** <fstart> <fstop> [points] (oscillator phase noise, E-140; run after **.hbosc**). Each reuses the same one-branch deck->control mechanism as **.hb** (E-162) / .sweep (E-146): in frontend/inp.c a top-level **.qps**/.**hbosc**/.**phasenoise** line is stripped of its leading . and appended to the post-parse control list, executing as the corresponding command once the circuit is built -- inheriting the full E-133/136/140 engines and their solver independence. Because the bridge preserves deck order, **.hbosc** (which finds the oscillator PSS + frequency) and **.phasenoise** (which reads it) compose, so a complete oscillator phase-noise run needs no .control block. A per-card boundary check (the char after the name is whitespace or end-of-line) keeps the cards distinct -- in particular **.hbosc** is NOT matched by the **.hb** branch, whose own boundary check rejects the trailing osc. Verified: the **.qps** dot-card produces a two-tone spectrum bit-for-bit identical to the qps command (examples/qps\_examples/verify\_qps.py); **.hbosc** + **.phasenoise** in a deck reproduce the command form's oscillation frequency and phase-noise spectrum exactly with order preserved (examples/phasenoise\_examples/verify\_phasenoise.py); both under Sparse and KLU. No regression to **.hb**/.**sweep**. | | [E-164](#) | (no ngspice source change) examples/rfpa\_examples/\* | Large-signal RF characterization of a real transistor -- a validation/example enhancement (no ngspice/openvaf-r source change) that closes the production-model validation loop (DC E-159, coverage E-160, small-signal E-161, now large-signal). It drives a common-emitter amplifier built from the bundled HICUM/L2 SiGe HBT (compiled in place, with the E-161 dynamic params  $t_0=10$ ps + 1fF caps) hard and extracts the large-signal RF figures of merit via TRANSIENT + Fourier/FFT: AM-AM gain compression (the power gain expands 17.8->20.7 dB then compresses -- the exponential-transconductance signature of a bipolar -- defining P1dB), harmonic distortion (the third harmonic follows the textbook 3:1 slope,  $HD_3 \sim A^3$ ), and two-tone third-order intermodulation (the IM3 products  $2f_1-f_2$  /  $2f_2-f_1$  at  $\sim 30$  dBc).

IIP3, extracted from the single tone as  $A/\sqrt{3HD3}$  (the two-tone IM3 is exactly 3x the single-tone HD3 for a third-order nonlinearity), is constant at  $\sim 0.134$  V across drive level, confirming the third-order model. A real finding: the frequency-domain harmonic-balance engines (hb/qps, E-134/136, which just gained netlist dot-cards in E-162/163) do NOT converge on this stiff, many-internal-node production model in an amplifier configuration -- hb returns error 103 at any drive level (even 1 mV) and with extra iterations/source-stepping, and the two-tone qps is prohibitively slow ( $>2$  min per point). Transient+FFT integrates reliably and is the robust route for large-signal RF on a full production compact model; HB/QPSS remain the tool of choice for lighter, well-conditioned circuits. Verified under both linear solvers (4 checks): transient small-signal gain matches .ac (7.744 vs 7.742), HD3 3:1 slope (62.9 vs 64), IIP3 constant (0.134 V, spread  $<2\%$ ), and gain compresses at high drive (7.74- $>5.97$ ). | | [E-165](#) | (no ngspice source change) examples/modelnoise\_examples/ | Production compact-model noise validation -- a validation/example enhancement (no ngspice/openvaf-r source change) exercising the last untested small-signal path, OSDI's .noise stamping of the models' own white\_noise / flicker\_noise sources, on production models compiled in place from the OpenVAF integration-test sources. BSIM4 (MOSFET): the output-noise spectral density  $S_v(f)$  of a common-source amplifier is compared to ngspice's BUILT-IN BSIM4 over the full band and matches to  $\sim 1.5\%$  everywhere, covering both the low-frequency  $1/f$  FLICKER region (amplitude density  $\sqrt{S_v} \sim 1/\sqrt{f}$ ) and the flat high-frequency THERMAL floor -- a stringent check since the flicker coefficients, thermal-noise model, and their bias dependence all have to line up. HICUM/L2 (SiGe HBT): no ngspice built-in, so validated against physics -- its default noise is WHITE (shot noise on the junction currents + thermal noise on the parasitic resistances, no flicker by default), a flat spectrum in contrast to the MOSFET's  $1/f$  rise; with a small source resistance so the intrinsic device noise dominates, the output-noise floor tracks the collector SHOT-noise line  $\sqrt{2qI_cRC^2}$  across two decades of bias current (within 6% at high bias where shot dominates) and scales as  $\sqrt{I_c}$ . Verified under BOTH the Sparse and KLU solvers (.noise works under both since E-113 fixed the KLU adjoint solve), 5 checks: BSIM4 spectrum matches built-in  $<4\%$ , BSIM4  $1/f$  flicker slope ( $S_v(1\text{Hz})/S_v(10\text{Hz}) \sim \sqrt{10}$ ), BSIM4 flat thermal floor, HICUM white noise, HICUM shot-noise tracking +  $\sqrt{I_c}$  scaling. Completes the production-model validation loop (DC E-159, coverage E-160, small-signal E-161, large-signal E-164, noise E-165). | | [E-169](#) | frontend/syntaxhl.c (new), include/ngspice/syntaxhl.h (new), frontend/commands.c, frontend/Makefile.am, main.c, examples/syntaxhl\_examples/\* | Interactive command-line syntax highlighting -- the interactive prompt now colors the command line AS IT IS TYPED. The command word is shown GREEN the moment it is a recognized command, RED when it can no longer become one, and left the terminal default while it is still a valid PREFIX being typed (so plo is neutral, plot green, plt red); numbers are yellow, "quoted strings" magenta, and -option flags cyan. The command word is classified exactly as the interpreter resolves it -- the same case-insensitive walk of the live command table cp\_coms that docommand() uses, plus the control keywords (if/while/foreach/begin/end/...) -- so the color always agrees with whether the word will actually run. Implementation: a pure engine cp\_highlight\_line() (frontend/syntaxhl.c) tokenizes a line and wraps each token in ANSI SGR color escapes; live coloring overrides GNU readline's rl\_redisplay\_function (installed by cp\_synhl\_init() from main.c right after rl\_outstream is set). The replacement redisplay redraws the colored line only when the prompt plus line fit on one terminal row and otherwise defers to readline's own rl\_redisplay(), so a wrapped line is never corrupted; the cursor is repositioned with a plain CR + CUF (cursor-forward) so mid-line editing is unaffected. Because that manual cursor positioning bypasses readline's internal position tracking, readline's own accept-line stops emitting the closing newline, so the Enter key is bound to a wrapper (cp\_synhl\_accept) that emits the newline itself when coloring is active before running the normal accept-line -- otherwise a command's output would begin on the input line (when coloring is off the wrapper is a no-op and readline handles the newline as usual). Gated three ways: it is on by default only at an interactive TTY (isatty(rl\_outstream)), is disabled by set no\_syntax\_highlight and by the NO\_COLOR environment convention, and never fires in batch (ngspice -b) or any piped/redirected session -- so simulation output is byte-for-byte unchanged (confirmed by the full example regression). Because as-you-type coloring needs the terminal in raw mode it requires a readline-enabled build, which the shipped binaries are (CI builds with --with-readline=yes and the committed binaries link libreadline). A new synhl <command line> command prints the colorized form of a line non-interactively -- useful for previewing and,

since it needs no terminal, what makes the coloring engine regression-testable in batch. Verified (examples/syntaxhl\_examples/verify\_syntaxhl.py, 11 checks): the engine via synhl in batch (valid command green, unknown red, valid prefix neutral, numbers/strings/options colored, case-insensitive) and the live as-you-type behavior driven through a pseudo-terminal (green/red/neutral on the fly, NO\_COLOR and a non-tty suppress all color); the live layer auto-skips on a build without readline. Solver-independent (front-end only). | | [E-170](#) | frontend/syntaxhl.c, include/ngspice/syntaxhl.h, examples/syntaxhl\_examples/\* | Semantic syntax highlighting -- extends E-169 from lexical to SEMANTIC coloring of the interactive line: it now checks whether the SIGNALS and EXPRESSIONS typed actually exist and parse. (1) A signal reference v(node)/i(source)/@device[param] is looked up read-only in the current plot (vec\_fromplot, no evaluation, no vector creation); it keeps the default color if it exists and is drawn RED if it does not. (2) Inside an otherwise-valid expression only the invalid signal reddens -- print v(a)+v(zzz) colors just v(zzz), including signals nested in functions (sqrt(v(a))-v(bad)). (3) A genuinely malformed expression argument of an expression command (plot/print/gnuplot/asciipLOT) reddens as a whole, tested with the real parser ft\_getpnames\_from\_string(FALSE) with its tree freed. (4) Error output is drawn RED at an interactive terminal: cp\_err is wrapped in a colorizing stream (funopen on BSD/macOS, fopencookie on Linux) that prepends the red SGR to each write; cp\_currerr is pointed at the wrapper too so ngspice's per-command stream reset cp\_ioreset() keeps it in place instead of restoring and closing it. A half-typed expression is NOT treated as an error and stays neutral -- a signal whose parenthesis has not closed (v(bP), an expression with unbalanced parens (v(a)+v(b) or one ending in an operator (v(a)+). KEY FIX: the expression parser's error handler PError writes straight to stderr, bypassing the cp\_err muting, so an early version spewed syntax error in line segment ... onto the prompt while typing; expr\_pares() now also mutes fd 2 (dup2 /dev/null) across the parse and restores it. Signal-validity is scoped to the unambiguous v()/i()/@ forms (bare words could be plot keywords like vs, functions like sqrt, or options -> false reds). Same three-way gating as E-169 (interactive TTY, set no\_syntax\_highlight, NO\_COLOR); the parser + vector lookups run per keystroke but are read-only and do NOT corrupt command execution (a subsequent let z=v(a)+v(b) still computes correctly). Verified (verify\_syntaxhl.py, 19 checks total): the E-169 lexical checks plus a semantic layer driven through a pseudo-terminal after simulating a small circuit -- valid signal not red, invalid signal red, invalid-in-expression reddens only the signal, malformed expression red as a whole, half-typed stays neutral with no parser-error spam, error output red, NO\_COLOR suppresses it. Solver-independent (front-end only). | | [E-171](#) | maths/KLU/klusmp.c, spicelib/analysis/pzan.c, examples/klupz\_examples/\* | KLU/Sparse solver audit -- KLU pole-zero fixed (complex determinant + pivot tolerance) -- a deep audit of the KLU<->SMP bridge found that pole-zero analysis under 'option klu' SILENTLY PRODUCED GARBAGE for any circuit with complex poles or zeros: a series RLC reported four bogus real poles (-100, -10, -0.89, 0) instead of its conjugate pair -5000 +/- j999987.5. Real-axis roots came out right (the mixed formula is accidentally correct for real pivots), which is why the existing single-real-pole RC regression never caught it. DEFECT 1 (spDeterminant\_KLU): the complex determinant built each pivot as the mixed quantity (1/(UxRs), UzRs) and took its complex reciprocal -- KLU's Udiag stores the ACTUAL pivots (its solve divides by them) while Sparse's Diag stores RECIPROCAL pivots (its solve multiplies), and the adaptation kept Sparse's reciprocal dance for the real part only. The real branch was worse: its product loop NEVER RAN (the loop index was left at N by the preceding permutation scan, so det was always +/-1.0), it divided instead of multiplying, and it never wrote piDeterminant (SMPcDProd consumed an uninitialized stack value). Both branches computed the permutation sign as #non-fixed-points/2, wrong for any cycle longer than 2 (a 3-cycle is EVEN but counted odd). Rewritten:  $\det(A) = \text{sign}(P)\text{sign}(Q)\text{prod}(U\text{diag}[k]R_s[k])$  with parity computed exactly by cycle decomposition (new PermutationParity helper); the KLU determinant now matches Sparse to ~14 digits at every Muller trial point. DEFECT 2 (SMPcReorder/SMPPreorder): pole-zero calls SMPcReorder with PivRel = 0.0; Sparse's spOrderAndFactor SANITIZES a non-positive threshold to its stored default, but the KLU branch assigned it straight to Common->tol, making KLU's partial pivoting accept an EXACTLY-ZERO diagonal as pivot ( $|\text{diag}| \geq 0_{\text{max}}$  always holds). At PZ's s=0 trial an inductor branch has a 0.0 diagonal (-sL), so the factorization returned KLU\_SINGULAR and PZ recorded a spurious root at the origin -- and the poisoned deflated Muller search never expanded past  $|s| \sim 10$ , yielding the garbage above. Both reorder entry points now mirror Sparse's sanitization (in-range

*PivRel still takes effect; DC/AC/tran pass the 1e-3 default and are byte-identical). With both root causes fixed, the E-113-era guard in pzan.c ('pole-zero finite-zero computation is not supported with option KLU' -- added when the zero search was seen to go spuriously singular, i.e. defect 2) is REMOVED; the finite-zero search now works under KLU and a genuine E\_SHORT reports the same 'input shorted' diagnostic as Sparse. The balanced/differential-output PZ guard remains (SMPcAddCol genuinely has no KLU branch). Verified (examples/klupz\_examples/verify\_klupz.py, 6 checks): full pole/zero root-set parity between the two solvers plus analytic anchors on a series RLC (conjugate pair), RC lowpass (old-good case, no regression), lead network (finite real zero), RC highpass (origin zero), RLC bandstop (PURELY IMAGINARY zeros  $+j1e6$ , the hardest case), and RLC bandpass (the finite-zero search formerly guarded off) -- all six identical across solvers. Full 134-example regression clean. Residual scope (solver-independent, ~1990 Muller-on-determinant fragility): a twin-T's deflated conjugate zero pair can still stall under KLU (4 of 6 roots found) while on the bandpass it is SPARSE that hits its iteration limit (KLU converges cleanly). | | [E-172](#) | maths/KLU/klusmp.c, spicelib/analysis/cktpzset.c, spicelib/analysis/pzan.c, examples/\_setup.py, examples/klupz\_examples/ | KLU balanced pole-zero + full partial pivoting -- closes E-171's two scope items; no analysis card remains Sparse-only under KLU. (1) BALANCED / DIFFERENTIAL-OUTPUT pole-zero ('pz in 0 a b vol', non-ground output reference) was guarded off as unsupported: the algorithm's change of variables needs the per-trial column fold  $\text{col}(b) += \text{col}(a)$  (SMPcAddCol), which had no KLU branch -- and KLU's CSC sparsity pattern is FIXED at COO->CSC conversion time, so the fold's fill-in cannot be created on the fly the way Sparse's spcCreateElement does. Fixed in two halves: CKTpzSetup reserves the UNION pattern before the conversion (every row of the solution column is also registered in the balance column via SMPmakeElt; duplicate (row,col) COO entries fold into a single CSC slot by the conversion's group labeling, so the reservation is safe), and SMPcAddCol gains a KLU branch that merge-walks the two columns' sorted row lists adding the complex values in place (with a defensive error if an addend row is missing, impossible after the reservation). The two pzan.c balanced-output guards are removed. (2) FULL PARTIAL PIVOTING for PZ factorizations: validating a fully-differential configuration exposed spurious far-field roots at  $|s| \sim 1e19.1e21$  (including a positive-real pole). An exact-rational (fraction-arithmetic) determinant of the very matrix KLU factored proved the factorization itself was numerically rotten at extreme  $|s|$ : at  $s=4.52e19$  the exact determinant was  $+4.36e11$  while the pivot product gave  $-2.05e12$  -- wrong sign AND magnitude. Root cause is architectural: KLU's ordering is fixed at klu\_analyze time (pattern-only AMD/BTF) while Sparse re-runs value-aware Markowitz ordering every PZ trial; PZ sweeps  $|s|$  across ~20 decades, and with the relaxed default  $\text{tol}=0.001$  the fixed ordering accepted catastrophically-cancelling pivots. KLU's only value-adaptive lever is the within-block partial pivoting, so the E-171 sanitization fallback is upgraded from 'keep current tolerance' to FULL partial pivoting ( $\text{tol}=1.0$ ) whenever the caller passes an out-of-range PivRel -- only pole-zero does (0.0); explicit in-range tolerances (DC/AC/tran 1e-3) are honored unchanged. This single change eliminated the far-field junk AND cured the twin-T conjugate-zero-pair stall recorded in E-171's scope (what looked like vintage-Muller noise-floor fragility under KLU was largely this factorization inaccuracy; KLU now finds all 6 twin-T roots with notch-zero real parts  $\sim 1e-12$ , an order CLOSER to the exact 0 than Sparse's  $\sim 1e-11$ ). Verified (verify\_klupz.py, 6 -> 9 checks): the E-171 six plus the differential RC bridge (balanced output, formerly 'not supported'), a balanced output with a complex pole pair, and the twin-T notch -- all root sets identical across solvers. The dual-solver harness's balanced-pz KLU-skip detection is removed from examples/\_setup.py (nothing remains Sparse-only). Full example regression 134/134. | | [E-173](#) | spicelib/analysis/cktpzeig.c (new), maths/dense/eig.c (new), maths/KLU/klusmp.c, spicelib/analysis/pzan.c, cktsopt.c, cktntask.c, cktdojob.c, include/ngspice/{smpdefs,optdefs,tskdefs,cktdefs}.h, two Makefile.am, examples/pzeig\_examples/ | Eigenvalue-based pole-zero ('options pzeig') -- a modern alternative root finder for .pz. The vintage driver hunts roots of  $\det(G+sC)$  with a Muller iteration on determinant values and is famously fragile (iteration limits -- 'giving up after 231 trials' on a plain RLC bandpass -- noise-floor stalls, chaotic search paths; E-171/172 fixed the KLU-side defects but the ~1990 algorithm itself stays delicate under BOTH solvers). The new opt-in method computes every pole and zero at once as a dense eigenvalue problem, reusing the existing CKTpzSetup/CKTpzLoad machinery so it applies unchanged to the poles and zeros configurations, to balanced/differential outputs, and to both linear solvers. Steps: (1) the PZ-configured MNA matrix is AFFINE in  $s$ ,  $A(s) = G + sC$ , so two loads ( $s=0$ ,  $s=1$ ) extract the*

dense pencil via a new solver-agnostic SMPdenseExtractReal (KLU: walk Ap/Ai/AxComplex; Sparse: walk FirstInCol through the Int->Ext translation maps); a third load at an arbitrary point VERIFIES affinity so a non-polynomial device falls back with a clear message instead of wrong roots. (2) C is structurally singular, so the pencil is solved by SHIFT-INVERT linearization: factor  $(G + \sigma C)$  ONCE at a non-root shift (tried from a small ladder of shifts, with the circuit's own sparse solver), form the dense  $M = (G + \sigma C)^{-1}C$  by  $n$  sparse solves; from  $(G + sC)v=0 \Leftrightarrow Mv = \mu v$  with  $\mu = 1/(\sigma - s)$ , every finite root is  $s = \sigma - 1/\mu$  and the pencil's infinite eigenvalues land harmlessly at  $\mu = 0$  (dropped by a noise-floor threshold  $64neps||M||$ ). (3) eigenvalues of the real dense M from a new self-contained eigensolver maths/dense/eig.c -- the classical balance / Hessenberg / Francis double-shift QR chain (EISPACK balanc/elmhes/hqr lineage, written fresh, no LAPACK dependency), unit-tested standalone against companion matrices, complex conjugate pairs, a 50x50 tridiagonal with known spectrum, and a badly-scaled balance case. Roots are emitted as the same PZtrial list the Muller driver produces, so PZpost and all downstream output are unchanged. Option plumbing follows the standard task-option path (OPT\_PZEIG -> TSKpzEig -> CKTpzEig flag -> dispatch in pzan.c). RESULTS: RLC bandpass -- same roots as Muller with NO iteration-limit warning; 10-section RC ladder -- all ten poles matching the analytic tridiagonal-eigenvalue formula  $s_k = -(2-2\cos((2k-1)\pi/21))/RC$  and Muller root-for-root; twin-T -- all 6 roots under both solvers; RLC bandstop -- purely imaginary zeros + j1e6 EXACT; balanced/differential output works; purely resistive circuits yield no roots and no crash. DEFAULT REMAINS MULLER (pzeig is opt-in). Dense  $O(n^2)$  memory /  $O(n^3)$  QR, capped at 2000 unknowns with a clear error above. Accuracy is dense-QR class (absolute error  $\sim \epsilon_{\text{ps}} * \text{dominant root magnitude}$ ): an exact origin-zero can print as  $\sim 1e-7$  when poles sit at  $1e6$  rad/s; Muller refines locally and can resolve wider per-root dynamic range, while pzeig never misses or invents a root -- the two methods agree to  $\geq 6$  digits on every battery circuit. Verified (verify\_pzeig.py, 13 checks: both solvers x both methods + analytic anchors + default-method check). Full example regression 136/136. || [E-174](#) | src/frontend/commands.c, examples/helpcmd\_examples/\* | **help all** crash fix (help string used as printf format) -- running help all (or help montecarlo) in the interactive shell crashed the shipped ngspice with 'Error: tvprintf failed / ERROR: fatal error in ngspice, exit(-1)'. Root cause: ngspice's help printer uses each command's help TEXT as a printf FORMAT string -- out\_printf(ccc[i]->co\_help, cp\_program) in com\_help.c (and the parallel com\_ahelp.c) -- passing only one argument (cp\_program, the program name, intended for a single %s as in the legitimate 'Report a %s bug.'). Any other '%' in a help string is an invalid/incomplete conversion specifier, so tvprintf fails and ngspice calls a fatal exit(-1). The montecarlo command's help (added in E-151) read '... reports the yield with a Wilson 95% CI ...'; the '%' (percent-space) is the invalid specifier. help all walks the command table alphabetically and died at montecarlo; help montecarlo crashed directly too. Present in BOTH command tables (spcp\_coms for ngspice, nutcp\_coms for nutmeg). Not X11/terminal/solver specific -- plain format-string handling, so it reproduced on every build; it was invisible in headless CI only because help all is an interactive command not exercised by the batch example decks. Fix: escape the literal percent as %% (printf convention) in both tables, so the line renders 'Wilson 95% CI' and help all completes. New regression guard examples/helpcmd\_examples/verify\_helpcmd.py with two layers: RUNTIME -- drives an interactive ngspice on a pseudo-terminal and asserts help, help all, and help montecarlo each run to completion (no crash, no 'tvprintf failed') and that the montecarlo line renders a literal '95% CI'; and a STATIC class-guard that scans commands.c and fails if any command help string carries a format hazard (a '%' that is not '%s' or '%%', or more than one '%s' since only one argument is passed), catching a future unescaped '%' even when that command is never exercised at runtime. Full example regression 137/137. || [E-175](#) | spicelib/analysis/dcpss.c, examples/rfconv\_examples/\* | RF-suite audit -- the dropped parametric term in every periodic small-signal analysis -- a full audit of the RF suite (.sp/Touchstone, PSS/PAC/pnoise/pxf, PSP, HB, QPSS + the two-tone family, hbosc/phasenoise) driven by analytic anchors and cross-analysis invariants. CLEAN: .sp + Touchstone (17 checks: series-R/shunt-C/3-port star S-parameters exact to 10 digits, the asymmetric-2-port S21/S12 column-order trap, reciprocity, RI/MA/DB + Y/Z (v1 normalization) + GHz round-trips through wrsnp/rdsnp, N-port 4-pairs-per-line layout); HB (analytic cubic: fundamental  $a1A + (3/4)a3A^3$  and  $H3 = (1/4)a3A^3$  exact to 7 digits, even harmonics at machine zero, Sparse/KLU identical); QPSS-HB/QPSS-tran (two-tone fundamentals,  $IM3 (3/4)a3A^3$  exact, commensurate aliasing guard fires correctly). THE BUG: every

LPTV small-signal conversion matrix scaled its reactive coupling by the COLUMN (input-sideband) frequency,  $H_{nm} = G_{\{n-m\}} + j\omega_m C_{\{n-m\}}$ . For a small signal riding a FROZEN periodic capacitance  $C(t)=dQ/dv$  the exact linearization is  $di = d/dt[C(t)dv] = C\dot{v} + C\dot{v}_{dot}$ , and the product rule collapses to the ROW frequency:  $H_{nm} = G_{\{n-m\}} + j\omega_n C_{\{n-m\}}$ . Using  $\omega_m$  silently DROPS THE PARAMETRIC-PUMPING TERM  $C\dot{v}$  -- *the very term that makes a pumped varactor convert. Every conversion sideband through a time-varying capacitance came out scaled by exactly  $\omega_{in}/\omega_{out}$ : on the audit circuit (1k -> OSDI varactor  $C(v)=c_0(1+\alpha v)$  pumped 1V@1MHz, 1mV probe at 250kHz) the +1/+2 sidebands were 3x/5x/7x/9x too small, with measured pre-fix/truth ratios matching the omega ratios to FIVE DIGITS (0.3333/0.2000/0.14286/0.11111) -- diagnosis confirmed beyond doubt. AFFECTED: pac, psp, pnoise, pxf, qpac, qpnoise, qpfx, and the phasenoise adjoint, for any circuit with pumped/nonlinear capacitance (varactors, junction and MOS capacitances -- every real mixer and oscillator). UNAFFECTED: LTI circuits (no off-diagonal C harmonics;  $\omega_n == \omega_m$  on the diagonal) -- exactly what every prior regression check used; the E-171 accidental-correctness pattern again. THE SUBTLETY: the harmonic-balance residual uses the SAME builders and must KEEP the column frequency -- its reactive current is the exact chain rule  $d/dt Q(v(t)) = C(v(t))\dot{v}(t)$  whose  $n$ -th harmonic is  $\sum_m C_{\{n-m\}}(j\omega_m V_m)$ . That is why HB and QPSS-HB always matched transient ground truth while the small-signal analyses did not, and why a blanket fix would have broken HB. FIX: a smallsig mode flag on pac\_build\_matrix and qp\_build\_matrix plus the two inline copies (the PAC adjoint and the phasenoise adjoint): smallsig=1 -> row frequency (restores the parametric term); smallsig=0 (HB/HBOSC/QPSS-HB residual + Jacobian) -> chain-rule column frequency, byte-for-byte unchanged. Regression safety proven directly: the linear-circuit PAC control produces IDENTICAL output pre/post fix, and HB/QPSS-HB outputs are byte-identical. Ground truth = plain transient + one-beat Fourier projection (independent of all conversion-matrix machinery); the fixed qpac matches all five sidebands within 2% (integrator error class). Also noted: cktspnoise.c is a dead fossil (#ifdef RFSPICE\_ never compiled). Verified (examples/rfconv\_examples/verify\_rfconv.py, 5 checks x both solvers: truth vs QPSS-HB unchanged, truth vs fixed qpac, pre-fix signature absent, HB + QPSS-HB analytic anchors). Full example regression 138/138. | | [E-176](#) | spicelib/analysis/dcpss.c, examples/\_setup.py, examples/pssdriven\_examples/\* | Driven-mode PSS shooting -- no frequency hunt, no breakpoint flood -- diagnosis of the E-175 audit's flagged residual ('PSS shooting robustness: the varcap PSS ran 17+ minutes without converging'). Instrumentation showed 9,627,161 accepted timepoints by  $t=2\mu s$  (~0.2 ps per step, zero LTE rejections): ngspice's PSS (the Lannutti shooting) is OSCILLATOR-oriented -- it hunts the fundamental frequency through mid-period forced breakpoints whose spacing is proportional to steady\_coeff. On driven circuits (everything the E-117..126 periodic small-signal stack runs on) this is doubly pathological: (1) at the standard decks' steady\_coeff=5e-6 the grid spacing is ~0.5 ps -- millions of steps per shooting cycle, and the convergence-tolerance knob doubles as grid density so tighter tolerance quadratically explodes runtime (even the default 1e-3 forces thousands of points/period -- why a linear RC .pss took ~4 minutes); (2) the frequency estimate can never settle EXACTLY on the source frequency, the cycle endpoint phase-drifts against the source, and the period residual FLOORS (~1e-4) -- shooting never converges and burns the full sc\_iter budget at flood speed. FIX: a DRIVEN mode, auto-detected when the circuit contains a time-varying independent source (SIN/PULSE/... V or I source, funcTGiven): the period is pinned to the exact source period (no estimation, no estimator grid; each cycle is one plain-transient period; varactor residual 2e-1 -> 8e-9 in 17 cycles / 0.3 s); the shooting-phase max step is clamped to  $T/psspoints$  so the orbit is integrated on the SAME discretization as the retained samples (validation caught this: without the clamp LTE grows steps toward  $T/2$  and shooting converges -- to 7e-9! -- onto the fixed point of that coarse discretization, several percent off the true orbit: retained swing 0.1651 vs analytic 0.15718; the old flood had been masking the effect by brute force); the post-FFT 'relaunch at the strongest spectral line' is disabled when driven (a rectifier-like circuit with a dominant 2nd harmonic must not retain the wrong period); the AUTONOMOUS oscillator path is untouched (byte-identical behavior verified on oscillator decks; DC-only-supply circuits never trigger detection). IMPACT: rc\_pss ~4 min -> 0.05 s with the retained fundamental EXACTLY 1000000 Hz (the old hunt returned 999999.8976); psp example 406 s -> 0.9 s; the rfpss battery (5 verifies, 61 checks) minutes-each -> <0.5 s each; rfanalyses 'several minutes even Sparse-only' -> 0.56 s; the KLU PSS pass 'prohibitively slow' -> 0.14 s. HARNESS: rfpss and rfanalyses removed*

from both REGRESSION\_EXCLUDE and SPARSE\_ONLY -- the whole RF periodic small-signal suite now runs in EVERY regression sweep under BOTH solvers; the direct .pac-on-pumped-varactor check became affordable, closing the E-175 loop on the direct PAC path (sidebands match transient ground truth within 1%). Verified (examples/pssdriven\_examples/verify\_pssdriven.py, 6 checks x both solvers: driven detection + exact retained frequency, retained swing ==  $|H|$  within 0.1%, retained-period self-consistency, direct .pac vs transient truth, autonomous non-trigger). Full example regression 145/145. | [E-177](#) | spicelib/analysis/dcpss.c, examples/pnoise\_examples/\* | pnoise noise-folding referee + the folded-flicker frequency fix -- closes the E-175 audit's last honesty-ledger gap: periodic noise analysis (pnoise/qnoise/phasenoise) was 'correct by construction, not correct by measurement' -- the folding of device noise through LPTV conversion had never been checked against anything independent. THE REFEREE (three independent layers, on a 1-node LPTV-conductance circuit  $G(t) = g_0 + g_1 \sin(w_0 t)$  chosen so conversion transfers are  $O(1)$ ): (1) a from-scratch Python conversion matrix,  $onoise(f) = \sum_k |Z_k(f)|^2 * S(|f + kf_0|)$  -- same physics, zero shared code; (2) a TRNOISE transient Monte-Carlo (no shared code OR theory; 198-segment Welch PSD, with the pumped/unpumped RATIO cancelling the TRNOISE amplitude convention); (3) the LTI (pnoise == .noise) and white limits. VERDICT ON THE WHITE PATH: measured-correct -- pnoise matched the referee to 6 digits at every frequency and the MC arbiter confirmed the ~1.86x folding ratio within statistical error; the adjoint, sideband sum and thermal densities are right. THE BUG THE REFEREE CAUGHT: the stationary sideband loops set the noise-evaluation frequency ONCE, to the output frequency (data.freq = freq outside the k loop). But the sideband-k adjoint carries noise ORIGINATING at  $|f + kf_0|$  -- a frequency-dependent source PSD (flicker  $1/f$ , noise\_table E-109) must be evaluated at the source-side frequency. The proof was digit-exact in both directions: pre-fix pnoise reproduced the referee's deliberately-wrong evaluate-at-f model DIGIT-FOR-DIGIT ( $1.114323e-14$  vs  $1.114323e-14$  at 1 kHz -- a 21% overestimate on this circuit, unbounded as  $f \ll f_0$ ); post-fix it reproduces the correct model digit-for-digit ( $9.175472e-15$ ). Why three generations of checks missed it: white noise is frequency-flat and LTI circuits have no  $k \neq 0$  transfer -- the THIRD occurrence of the accidental-correctness pattern (E-171 determinant, E-175 parametric term). FIXED in all three stationary loops: pnoise (data.freq =  $|freq + kf_0|$  per sideband), qnoise ( $|f_{in} + k_1 f_1 + k_2 f_2|$  per harmonic), and phasenoise -- where it matters most: folded far-sideband blocks near a carrier were evaluated at the tiny OFFSET frequency, inflating the flicker ( $1/f^3$ ) region. CYCLO SCOPE: the cyclo mode's time-domain identity  $onoise = (1/P) \sum_s S(t_s) |A_s|^2$  treats the source PSD as frequency-flat across the folding span -- exact for modulated-white noise and for flicker on conversion-free circuits (the shipped E-125/126 use cases), not for flicker folded through  $k \neq 0$  conversion (cannot be collapsed into that product with the aggregate device-noise API); documented at both cyclo sites, with the (now exact) stationary mode as the right tool when folded flicker matters. Verified (examples/pnoise\_examples/verify\_pnoise\_examples.py, 5 checks x both solvers, ~2 s thanks to E-176): white folding vs referee ( $\leq 0.1\%$ ), flicker folding vs referee ( $\leq 0.5\%$ , sample-0 bias read from the retained orbit), pre-fix signature absent, pump/no-pump ratio == referee ratio, LTI limit == .noise. The rfpss/qpss/rccyclo suites pass unchanged (conversion-free or white -- provably unaffected). Full example regression 146/146. | [E-178](#) | spicelib/analysis/dcpss.c, spicelib/devices/dio/dionoise.c, examples/cyclo\_examples/, examples/rfpss\_examples/, examples/qpss\_examples/\* | exact separable cyclostationary folding + the HB DC-source fix -- removes the cyclo mode's frequency-flat limitation E-177 documented. For the separable model  $S(t,f) = m(t)^2 g(f)$ ,  $onoise(f) = \sum_g \sum_q |B_q(g)|^2 * g_g(|f + qf_0|)$  with  $B_q(g) = (1/P) \sum_s m_g(t_s) dA_s(g) e^{+jq \theta_s}$  and  $A_s = \sum_k \Psi_k e^{-jk \theta_s}$ . Per-generator amplitudes recovered with NO device-API change: the noise-summary machinery (prtSummary/outpVector, one density per generator; family totals excluded by the name-prefix rule) plus load POLARIZATION against a fixed reference load  $R = \Psi_0$  -- five DEVnoise sweeps per orbit sample ( $A+R, A+jR, R$ ) give  $c_{g,s} = \sqrt{S_g(t_s)} dA_s(g)$  up to a constant phase that cancels in  $|B_q|^2$ . Spectral shapes are MEASURED pointwise at the folded frequencies (no  $1/f^{EF}$  assumption -- noise\_table works) at several probe biases; flat generators keep the original time-domain identity (exact for ANY quadratic form incl. NevalSrc2-correlated pairs); stationary limit == the E-177 sum, flat limit == the old identity (Parseval). Ported 2-D to qnoise ... cyclo ( $B_{\{q_1,q_2\}}$  at  $|f + q_1 f_1 + q_2 f_2|$ ). THE PHYSICS: 'flicker sees  $\wedge^2$ , white sees  $\langle m^2 \rangle$ ' -- only the envelope's DC feeds the  $1/f$  band; the old identity was 23% high ( $8/\pi^2$ ) on  $|\sin|$ -modulated flicker and 34% high on the conversion circuit; the new

sweep lands on a from-scratch referee to  $\leq 3e-4$  (the old binary reproduced the deliberately-flat model to  $6e-5$  -- the error was the model, not the code). BONUS BUG (4th accidental-correctness occurrence): the cross-machinery check (same circuit through the QPSS-HB orbit) came out exactly 4x -- BOTH HB Newtons (hb E-134, qpss ... hb E-136) double-subtracted the DC sources (the settle-mode rhs folded into I\_R already carries them; *-lambdaIs subtracts them again*): *every DC bias voltage converged to exactly 2x, silently corrupting all bias-dependent noise and conversion (flicker  $\sim I^{AF}$  off by  $2^{AF}$ , shot 2x) while AC content and every AC-driven validation stayed exact. Fixed by adding the unscaled DC source block back (net DC drive exactly -lambdaIs\_DC);* qpnoise cyclo now agrees with pnoise cyclo DIGIT-FOR-DIGIT across the two orbit machineries. rcyclo's flicker law updated to the exact  $\langle |I| \rangle^2 = (8/\pi^2) \langle I^2 \rangle$  (pinned to  $1e-4$ ). 6 checks x both solvers; regression 147/147. FOLLOW-UP (same enhancement): the E-139 hard-pumped-diode deck exploded the cyclo path ( $\sim 6.5e1 \text{ V}^2/\text{Hz}$ ) -- dionoise.c computes its three sidewall generators only when DIOresistSWGIVEN but the prtSummary block copies ALL DIONSRCs slots, so without a sidewall model the summary vector carried uninitialized stack garbage (latent in stock ngspice for decades; E-178 is the first consumer of per-generator densities). Fixed at the device (unwritten slots zeroed, dio/dionoise.c) and hardened in both cyclo consumers (finite non-negative PSD guards; garbage probe ratios fall back to the flat path). verify\_qpnoise's 'cyclo > 2x stationary' diode expectation was an artifact of the garbage slots + doubled HB DC bias; replaced with a closed-form torus-average referee (resistive network  $\Rightarrow$  memoryless LPTV, instantaneous adjoints) matched to  $\sim 1\%$ . || [E-179](#) | spicelib/analysis/tfana.c, spicelib/analysis/cktsens.c, frontend/com\_measure2.c, examples/stdaudit\_examples/\* | standard-analyses audit: referee battery + three fixes. The 'Standard analyses (analog)' gap-table rows are 1990s SPICE3 code whose VALUES had never been physics-checked. FIX 1 (tfana.c, inherited from Berkeley SPICE3 1988): .tf i(vm) vin output impedance was ALWAYS  $1e20$  -- the output-impedance solve forces 1V on the sense branch and inverts the branch current as  $1/\text{MAX}(1e-20, \text{rhs})$ , but that current is NEGATIVE for every passive network (the input-impedance path right above divides by -rhs); fixed with the same sign + fabs guard; feedback-amp referee exact ( $1000.990 == \text{RL} + \text{node Thevenin}$ ). FIX 2 (cktsens.c): KLU AC sensitivity silently truncated to ONE frequency point -- the E-114 KLU complex-conversion block inside the frequency loop reused i (the OUTER loop variable) for its DEVmaxnum walk, so after the first point  $i = \text{DEVmaxnum}$  ended the sweep; the surviving point was correct, which is why E-62's single-point check passed. Fixed with a local index (dead second block hardened too); full sweep  $\Rightarrow$  analytic  $dV/dC = -j\omega R / (1 + j\omega RC)^2$  to 6 digits, both solvers. FIX 3 (com\_measure2.c): .meas DERIV{ATIVE} was parsed but evaluation fell into an explicit 'currently not supported' stub (empty #if 0 measure\_deriv() placeholder) -- implemented: 3-point Lagrange-quadratic derivative on the nonuniform time grid, AT= and WHEN forms sharing FIND's parse/locate flow, verified vs the analytic sine slope to 5 digits; DERIVATIVE/INTEGRAL long spellings accepted (only DERIV/INTEG matched); pre-existing -Wmissing-prototypes warning silenced (str\_has\_arith\_char2 static). MEASURED-CORRECT: .disto HD2/HD3  $\Rightarrow$  analytic diode Volterra kernels  $\leq 0.06\%$  incl. a frequency-dependent load (harmonics at  $Z(2\omega)/Z(3\omega)$  -- no E-177-style frequency bug), SIM2 intermods ( $f_1+f_2, f_1-f_2, 2f_1-f_2$ ) incl. cascades, exact DISTOF1/DISTOF2 scaling, junction-cap harmonics  $\Rightarrow$  Harmonic Balance to  $\sim 6$  digits (two independent engines); .noise onoise\_total^2  $\Rightarrow$  band analytic ( $kT/C$ )  $0.06\%$  + flicker log-integral 6 digits; .sens DC  $\Rightarrow$  central finite differences at a nonlinear OP (5-6 digits, model params incl.); nonlinear-OP .pz works via the driving-point form (the E-62 'quirk' was a bias source shorting the injection node -- ngspice's refusal is correct); tran/op/meas battery exact. 8 checks x both solvers; regression 148/148. || [E-180](#) | frontend/com\_checkpoint.c, examples/checkpoint\_examples/\* | checkpoint/restart under KLU (the last Sparse-only feature falls). E-131 guarded savestate/loadstate off under KLU, recording the reason as 'KLU factorization objects absent on the restore path' -- a mis-diagnosis. REAL cause: .option klu lives in the task (TSKkluMODE) and reaches the circuit only in CKTdoJob at analysis dispatch, but loadstate calls CKTsetup DIRECTLY first, so the matrix was built Sparse (SMPkluMatrix NULL); the resume dispatch then set  $\text{ckt} \rightarrow \text{CKTkluMODE} = 1$  without re-setup (by design -- it must preserve the restored state), and the first NlIter dereferenced the NULL SMPkluMatrix (KLUloadDiag-Gmin store; reproduced EXC\_BAD\_ACCESS +0x90). savestate was never broken (the checkpoint file holds only solver-agnostic vectors). FIX: com\_loadstate copies the task's solver selection and the E-152 KLU knobs (ordering/scale/BTF/memgrow) into the circuit BEFORE CKTsetup, exactly CKTdoJob's

block; both guards removed. BONUS: cross-solver restore -- all four save x load combinations (fresh processes) continue exactly on the uninterrupted trajectory ( $\leq 1e-9$  at  $t=3.5\text{ms}$  of a  $4\text{ms}$  diode+RC run). `verify_checkpoint`'s 'KLU rejected' check replaced by the 4-combo matrix (22/22); solver-notes and gap-analysis tables updated. Nothing in the simulator is Sparse-only any more (E-172 closed the last analysis, E-180 the last feature). Regression 148/148. |

*(E-25's `simpparam` exposure lives in the OSDI callback table populated at load time; its diff rides inside the `osdi_load.c/callbacks` changes.)*

*Build-warning cleanup (post-E-127): a full rebuild under the repo's `-Wconversion` flags surfaced latent warnings. `configure.ac` had `-s` (a strip flag) in the compile CFLAGS, so clang warned "argument unused during compilation: '-s'" on every source file (~1526 warnings) — removed (binaries are stripped separately by the fold and CI). `-Wconversion` also flagged a real latent bug in `maths/ni/nipzmeth.c` (`b = 0.0` was an assignment where `b == 0.0` was intended) — fixed — and benign `double→bool` conversions in `maths/cmaths/cmath4.c` — made explicit. Full-build warnings dropped 1903 → 394; the remainder are `-Wconversion` warnings in third-party SuiteSparse (KLU) and CMC device models (HiSIM), which are left to their upstream sources.*

Every entry above is guarded by a verify script under [examples/](#) and was regression-checked against the full example arsenal, the compiler's integration suite, and the VA\_TEST industry-model corpus at the time it landed.